

Financial Hedging and Firm Performance: Evidence from Cross-border Mergers and Acquisitions

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Abstract

Using a sample of 1,369 cross-border acquisitions announced by Standard & Poor's 1500 firms between 2000 and 2014, we find strong evidence that derivatives users experience higher announcement returns than non-users, which translates into a US\$ 193.7 million shareholder gain for an average-sized acquirer. In addition, we find that acquirers with hedging programmes have higher deal completion probabilities, longer deal completion times, and better long-term post-deal performance. We confirm our findings after employing an

The authors would like to thank the two anonymous referees, the journal editor, John Doukas, Eric de Bodt, Chris Brooks, Tao Chen; Zhiyao Chen, Michael Clements, Pasquale Della Corte, Espen Eckbo, Zhan Feng, Ranko Jelic, Wei-Ming Lee, Lalitha Naveen, Annette Poulsen, Karin Thorburn, Baolian Wang, Ke Yang, and seminar participants at the ICMA Centre of Henley Business School, Central Washington University; the Second Young Finance Scholars' Conference at the University of Sussex; the IFABS 2015 Corporate Finance Conference at Oxford University; the 15th Finance, Risk and Accounting Perspectives Conference; the 2015 Financial Management Association Annual Conference; the 2015 Paris Financial Management Conference the Fourth ECCCS Workshop on Governance and Control; and the BAFA Annual Conference 2016 for their insightful and constructive comments. The authors also thank Anh To and Norika Kobayashi for their research assistance. Financial support from the ICMA Centre is gratefully acknowledged.

extensive array of models to address potential endogeneity. Overall, our results provide new insights into a link between corporate financial hedging and firm performance.

Keywords: *cross-border M&As, risk management, financial derivatives*

JEL classification: *F31, G13, G32, G34*

1. Introduction

Financial derivatives have been widely used as risk management instruments by companies around the world.¹ Despite the popularity of financial derivatives among corporations and the theoretical groundwork in support of corporate financial hedging (e.g., Mayers and Smith, 1982; Stulz, 1984; Smith and Stulz, 1985; Leland, 1998; Froot *et al.*, 1993; DeMarzo and Duffie, 1995), empirical findings on the relationship between corporate financial hedging and firm value are still mixed. For example, Allayannis *et al.* (2001) find that the use of FCDs is positively correlated with firm value in a large sample of US non-financial corporations with foreign currency risk exposures. Carter *et al.* (2006) report a significantly positive hedging premium among a sample of US airline firms that engage in jet fuel hedging. However, Guay and Kothari (2003) show that the cash flows generated by hedging are modest and cannot account for a large increase in firm value. Jin and Jorion (2006) study a sample of oil and gas producers and find no significant impact of financial hedging on firm value.

In addition to the mixed findings on the relationship between hedging and firm value, empirical studies that examine the effect of hedging on firm operating activities have been surprisingly lacking. In this paper, we fill this gap by studying an important form of firm operations that is well known for changing a company's financial risk exposure: cross-border mergers and acquisitions (M&As). By examining the impact of hedging on deal performance, we seek to contribute to the literature on whether and how risk management affects firm performance. Cross-border M&A activities have been on the rise around the world over the last two decades as a result of deregulation and the globalisation of product and capital markets.² Compared to domestic M&As, cross-border M&As involve additional risk

¹ A 1998 Wharton survey of financial risk management by US non-financial firms (Bodnar *et al.*, 1998) finds that 41.5% of respondents use foreign currency derivatives (FCDs) and 38% firms use interest rate (IR) derivatives. An International Swaps Derivatives Association (ISDA) survey in 2009 shows that 94% of the world's 500 largest companies use derivatives to manage their business and financial risks, of which 88% use derivatives to manage foreign currency risk. According to the statistics released by the Bank for International Settlements, the notional value of outstanding IR and FCDs held by global non-financial customers was US\$ 15.7 trillion and US\$ 9.1 trillion, respectively, at the end of June 2014.

² Thomson Reuters reports that global cross-border M&As reached a peak of US\$ 1.8 trillion in 2007, accounting for 44.8% of the overall M&A volume. After the global financial crisis, the value of global cross-border M&As hit bottom in 2009 and then gradually recovered to US\$ 1.3 trillion in 2014, accounting for 36.9% of the overall M&A volume.

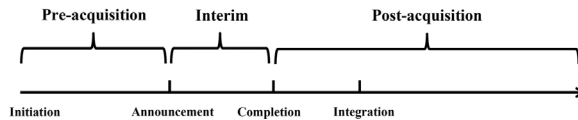


Fig. 1. Financial risks along a cross-border M&A timeline

This figure presents a typical cross-border M&A timeline and the associated financial risks at different phases of the deal. We define the pre-acquisition period to be between the deal initiation date and the announcement date. During this period, acquirers and targets privately negotiate with each other. Next, we define the interim period to be between the announcement date and the deal completion date. Finally, we define the post-acquisition period as the period after the deal completion date, which can be further divided into the integration stage and the post-integration stage.

- Pre-acquisition period: Evaluating the target's financial risk
- Interim period: Transaction risk
 - Transaction risk refers to the uncertainty that the US dollar value of a target's price and the external financing costs could change with the FX rate and the interest rate between deal initiation and deal completion.
- Post-acquisition period: Integration risk, balance sheet risk, cash flow risk
 - Integration risk refers to the uncertainty for an acquirer when designing the new financial risk management programmes of the combined entity during the integration period.
 - Balance sheet risk refers to the uncertainty about the US dollar value of acquired assets or liabilities due to the movement of FX rates.
 - Cash flow risk refers to the uncertainty about the US dollar value of future operating cash flows due to the movement of FX rates.

elements due to differences in culture, geography, capital market development, accounting rules, and regulations between the acquirer and the target countries. More importantly, taking over a foreign target significantly changes the acquirer's financial risk exposure. As shown in Figure 1, the timeline of M&A transactions can be divided into three stages: the pre-acquisition period (between the start of private negotiation and the deal announcement), the interim period (between the deal announcement and completion), and the post-acquisition period (after deal completion).³ An acquirer in a cross-border M&A encounters deal-related risk exposures at different stages. During the pre-acquisition stage, both the acquirer and the target encounter valuation fluctuation due to exchange rate movements, which could change the target's attractiveness to the acquirer. During the interim stage, an acquirer is exposed to new foreign exchange (FX) risk if the payment is denominated in the target nation's currency. In addition, if an acquisition requires external financing, the acquirer is exposed to IR risk, which affects the deal financing costs. During the post-acquisition stage, an acquirer faces greater FX exposures as a result of acquiring the foreign target and increased foreign operations.

There are four main advantages to studying the impact of hedging on firm performance through cross-border M&As. First, this sample naturally excludes firms without FX exposure. A firm that does not have significant financial risk exposure may choose not to

³ We follow Ahern and Sosyura (2014) to define these three periods.

hedge due to the lack of hedging benefits. If a sample includes firms of this nature, then the empirical tests to examine the impact of hedging on firm performance could be compromised because they do not serve as an effective control group. By focusing on firms experiencing a significant increase in their risk exposure, we are able to conduct more effective tests to identify the potential hedging benefits. Second, this sample allows us to adopt the standard M&A event study approach, which mitigates the reverse causality concern that firms with better performance are more likely to hedge rather than hedging leading to improved deal outcomes. By examining acquirers' established derivatives contracts before their deal announcements, when firm performance is measured, this approach alleviates the reverse causality concern because it is unlikely that the M&A announcement returns are responsible for the establishment of hedging programmes in acquiring firms. Third, the event study approach allows us to examine the impact of hedging on firm performance measured by stock market reaction and firm operating performance, which offer an alternative to the firm value measure, Tobin's Q, adopted by most risk management studies. Acquirer announcement returns directly measure the acquirer shareholder wealth effect associated with hedging. Finally, being the first to directly study the effect of financial risk management on a specific type of firm operation, our results can shed light on the channels through which financial hedging improves firm value.

Our sample includes 1,369 cross-border M&A deals announced by Standard & Poor's (S&P) 1500 firms between 2000 and 2014.⁴ The S&P 1500 index covers 90% of the US stock market capitalisation and includes firms of various sizes (S&P 500 Large Cap, S&P 400 Mid Cap, and S&P 600 Small Cap) and industries. For each deal, we manually collect the acquirer's derivatives data reported in its 10-K report prior to the corresponding deal announcement date, following the derivatives data collection procedure of Allayannis and Weston (2001) and Bartram *et al.* (2011).⁵

Our sample firms make foreign acquisitions in over 35 countries. More than 72% of these acquirers use either FCDs or IR derivatives (IRDs) to manage firm financial risks. About 63% of acquirers use FCDs and more than 37% of acquirers have FCD positions in the currencies of target nations. Our univariate tests show that derivatives users are better than non-users in terms of both short- and long-term performance. We also find that derivatives users and non-users differ significantly in terms of size, leverage, cash ratio, stock return momentum and volatility, as well as return on assets (ROA). In addition, the use of derivatives is more prevalent among acquirers that take over large targets and use less equity as the method of payment. These differences emphasise the importance of multivariate tests.

⁴Empirical research on corporate financial hedging after 2000 is scant due to the change in accounting rules on derivatives information disclosure in firms' annual financial reports. See online Appendix A for further discussion on the changes in these accounting rules.

⁵In online Appendix B, we present a few excerpts from acquirers' 10-K reports. These statements provide direct evidence that acquirers use derivatives to actively manage the financial risks associated with their cross-border M&As. Under current financial accounting reporting standards, it is not mandatory for US public firms to disclose their hedging position details. Therefore, our hedging proxy variables can only represent firm-level financial hedging, not deal-specific hedging. A portion of firms in our sample disclose M&A deal-specific hedging activities in their 10-K reports.

Our multivariate tests confirm that financial hedging is positively associated with acquirer cumulative abnormal returns (CARs) during the announcement period. Hedging with currency derivatives is associated with an average 1.0% improvement in acquirer CARs, which is equivalent to an increase of US\$ 193.7 million in shareholder value for an average-sized acquirer. We also find that financial hedging has an effect on other aspects of deal outcomes. First, deals made by firms with financial hedging programmes have a higher completion probability than deals by firms without such programmes. Next, we show that it takes FX derivatives users an average of 32 days longer to complete a deal than non-users. Our observation is consistent with the interpretation that waiting costs for derivatives users are lower than for non-users during the deal negotiation phase.⁶ We also investigate whether the short-run superior performance of derivatives users extends to the long run. In terms of operating performance, we find that the change in abnormal ROA (AROA) for FCD users is, on average, 0.7% higher than for non-users over a 3-year period after M&A deals. The improved long-term performance related to hedging is also confirmed by measures of stock returns. Over a 3-year horizon, FCD users experience 0.4% higher buy-and-hold abnormal stock returns (BHARs) than non-users. Overall, we find strong evidence that financial hedging is related to improved deal performance for acquirers engaged in cross-border M&As.

A major concern for the interpretation of our empirical results is that the decision to establish a financial hedging programme is not random. It could depend on omitted firm characteristics variables that could independently affect M&A outcomes. To address this concern, we employ three different methods. First, we adopt a special form of the Heckman (1978) selection model: an endogenous treatment effect model with two-step consistent estimates (Wooldridge, 2010). This approach helps us estimate the ‘treatment effect’, that the superior deal performance of derivatives users over non-users is caused by hedging instead of other unobserved variables. We adopt two instrument variables from Geczy *et al.* (1997), who find that the number of analysts following a firm and the ratio of foreign sales over total sales can explain why firms use FCDs. Second, we follow Gormley and Matsa (2014) and estimate a triple interaction fixed effects model, that is, $\text{Year} \times \text{Acquirer Industry} \times \text{Target Industry}$, to control for unobserved heterogeneity. Finally, we conduct a propensity score matching (PSM) analysis. We pair each acquisition announced by a derivatives user in our sample with a group of similar deals announced by non-users using observed deal and firm characteristics. Then we compare the deal performance between these two samples. Our findings on the relationship between financial hedging and deal performance remain robust in these three tests.

To further check on alternative explanations for our empirical results, we conduct a battery of robustness tests. First, since acquiring firms with high-quality management teams, extensive foreign operating exposure, and rich acquisition experience are more likely to use financial derivatives and have better announcement returns, we verify that the positive effect of financial hedging on acquirer announcement returns remains robust after including all these additional control variables. Second, we add commodity derivatives (CDs) as a control variable in our tests, because CDs are related to firm characteristics that determine a firm’s risk management policy but are not directly related to the financial risks associated with cross-border M&A deals. We find that FCD and IRD

⁶Offenberg and Pirinsky (2015) suggest that a deal’s completion time is determined by the trade-off between the costs and benefits of waiting.

variables remain significantly positively related to acquirer announcement returns but the CD variable does not. This finding is inconsistent with the notion that derivatives hedging only signals unobserved acquirer characteristics. Finally, by dividing our sample periods into a high exchange rate volatility regime and a low-volatility regime, we show that the positive effect of FX hedging on acquirer announcement returns is more significant when exchange rate volatility is high, suggesting that derivatives are more beneficial to acquirers under volatile market conditions. Related to this, we confirm that the implied volatilities of at-the-money (ATM) options written on acquirer stocks at deal announcement are lower for derivatives users than for non-users.

The paper proceeds as follows. Section 2 reviews the related literature on risk management and cross-border M&As. Section 3 develops the main test hypotheses and predictions. Section 4 describes the sample, variable definitions, and summary statistics. Section 5 reports our main empirical results and section 6 presents robustness checks and additional discussions. Finally, section 7 concludes the paper.

2. Related Literature

2.1. Financial hedging and firm value

Within the classic framework of Modigliani and Miller (1958) risk management has no impact on firm value and hedging is irrelevant if investors can engage in ‘homemade’ hedging. However, corporate risk management becomes valuable due to the existence of market imperfections in practice. Several motivations of corporate financial hedging have been developed in previous literature, including reductions in financial distress costs (Mayers and Smith, 1982; Stulz, 1984; Smith and Stulz, 1985), the alleviation of information asymmetry (DeMarzo and Duffe, 1995), the mitigation of agency costs associated with underinvestment and risk shifting (Stulz, 1984; Froot *et al.*, 1993; Leland, 1998), reductions in the cost of debt (Chen and King, 2014), reductions in the cost of equity (Gay *et al.*, 2011), and the alleviation of effective tax payments (Graham and Rogers, 2002). Survey results also suggest that academics commonly support the view that corporate financial hedging generally helps firms manage risks efficiently and increases shareholder value.⁷

Despite the theoretical groundwork, empirical evidence on the relationship between financial hedging and firm value remains mixed. Geczy *et al.* (1997) examine a sample of US non-financial firms and find that hedging reduces firms’ cash flow fluctuations. Allayannis *et al.* (2001) identify a hedging premium of about 5% of firm value for a sample of US non-financial firms. Carter *et al.* (2006) report a higher hedging premium of 10% based on a sample of firms in the airline industry. Using international data,

⁷ In March 2004, the ISDA conducted a survey of finance professors at the top 50 business schools worldwide to solicit their opinions on corporate financial hedging, as well as on the impact of derivatives on the global financial system. A total of 84 professors from 42 institutions provided responses. When asked to rate the statement ‘Managing financial risk more effectively is a way for companies to build shareholder value’, 44% strongly agreed, 47% agreed, 7% somewhat agreed, and only 2% somewhat disagreed. When asked whether ‘derivatives help companies manage financial risk more efficiently’, 49% strongly agreed, 43% agreed, 8% somewhat agreed, and no participant disagreed.

Bartram *et al.* (2011) find strong evidence that hedging firms have significantly lower cash flow volatilities, total risk and systematic risk. On the other hand, Tufano (1996) studies the hedging policy of a sample of gold mining companies but fails to find a significant relationship between risk management and firm value. Guay and Kothari (2003) conclude that corporate derivatives positions are generally far too small to account for the valuation premium observed by Allayannis *et al.* (2001). Jin and Jorion (2006) conclude that financial hedging has no significant effect on the market value of a sample of US oil and gas producers. Similarly, Brown *et al.* (2006) confirm that selective hedging does not lead to better operating or stock performance among a sample of US gold mining firms. One potential explanation for these mixed empirical findings is self-selection bias. The FX exposures of US corporations are determined by their foreign operations (Allayannis and Ihrig, 2001) and companies with variable risk exposures obtain different hedging benefits in terms of value improvement (Pritamani *et al.*, 2004).

A recent strand of literature studies the potential channels through which firms benefit from financial hedging. Campello *et al.* (2011) find that derivatives users receive more favourable financing terms in their loan agreements than non-users. Graham and Rogers (2002) report that hedging helps increase debt capacity, leading to an average 1.1% increase in firm value. Chen and King (2014) confirm that the bond yield spread for derivatives users is lower than for non-users. Gay *et al.* (2011) report that firms using derivatives have a lower cost of equity than non-users. Pérez-González and Yun (2013) use the innovation of weather derivatives as a natural experiment and find that the introduction of such derivatives leads to higher leverage, more investment, and ultimately higher firm value for a sample of electric and gas utility firms with weather risk exposures.

2.2. Financial hedging and M&As

Not only do M&As have a huge impact on an acquirer's future operations and growth, but they also significantly change its risk profile. For example, Furfine and Rosen (2011) find that M&As increase the average default risk of the acquiring firms. Duchin and Schmidt (2013) report that acquisitions completed during US merger waves have higher uncertainty and information asymmetry than those occurring outside of merger waves, leading to poorer deal outcomes. Bhagwat *et al.* (2016) suggest that M&A activities decrease when stock market volatility increases, because the higher interim period risk makes deals less attractive to both potential acquirers and targets. Bhagwat and Dam (2014) provide evidence that acquirers bear more interim period risk than targets.

Cross-border M&As have received increased attention in recent studies because acquirers in these deals experience significant FX risk change. Moeller and Schlingemann (2005) confirm that cross-border M&As increase acquirers' foreign currency exposures. They also report that cross-border acquirers have poorer stock and operating performance than acquirers in domestic deals. To study how firms respond to exchange rate shocks in making investment decisions, Lin *et al.* (2014) examine around 1,000 cross-border M&As and find that acquirers in countries experiencing strong currency appreciation have higher abnormal announcement returns. They further conclude that this wealth effect is more significant for acquirers with strong corporate governance. Bartram *et al.* (2013) examine a sample of US firms acquiring foreign targets and conclude that these acquisitions reduce the acquirers' currency risk exposures if they have prior presence in the target's countries. Lin *et al.* (2009) conclude that,

because financial derivatives users suffer less information asymmetry than non-users, they tend to have better long-run stock performance than non-users after cross-border M&As.

Despite the wide range of studies on market uncertainties and M&As, empirical research is still scarce on whether acquirers use financial derivatives to manage their deal-related risk and, if they do, how successful they are in this effort. Our paper is among the first to shed light in this area.

3. Hypotheses and Empirical Predictions

In this paper, we examine different deal outcomes between financial derivatives users and non-users in cross-border M&As. Employing a standard M&A event study approach, we develop and test four hypotheses.

H1: *Acquirer CARs around cross-border M&A announcements are higher for financial derivatives users than for non-users.*

Financial hedging is beneficial for an acquirer over the timeline of a cross-border M&A transaction. During the pre-acquisition period, an acquirer with existing risk management programmes has better ability and lower costs in evaluating the financial risks associated with the potential deal. At the deal's announcement, financial hedging sends investors a positive signal of an acquirer's financial risk management expertise and mitigates the information asymmetry problem between firm managers and outside investors (DeMarzo and Duffe, 1995). During the interim period, regardless of the payment method, financial hedging reduces the acquirer's FX risk exposures associated with deal payments. If an acquirer needs external financing, then IRDs help it manage the IR risk exposure. Financial hedging is also directly associated with lower external financial costs (e.g., Campello *et al.*, 2011; Chen and King, 2014). After deal completion, an acquirer with hedging experience can design more effective risk management strategies for the combined entity in the integration process. During the post-acquisition period, an acquirer's balance sheet FX risk will increase, because the acquired assets and liabilities are denominated in foreign currencies. The FX risk exposures of an acquirer's future operating cash flows will also increase after the foreign target is acquired. An acquirer that uses derivatives can manage both balance sheet and operating cash flow FX risks more efficiently than an acquirer that does not, leading to superior market reaction at the deal announcement.

The benefits of financial hedging associated with acquirer announcement returns can be categorised in the following way. First, financial hedging can effectively reduce both the total and systematic risk of a company (Bartram *et al.*, 2011). Acquiring a foreign target greatly changes a company's financial risk exposure and both FCDs and IRDs reduce deal-related financial risk. Second, hedging experience helps an acquirer evaluate the financial risk associated with the M&A deal, which can help choose a better target. An acquirer with hedging capacity could also be in a better negotiation position to obtain better deal terms. Third, financial hedging sends outside investors a positive signal about a firm's international operations experience and managerial ability, which can help them assess deal quality more precisely and reduce information asymmetry. Fourth, financial hedging reduces deal transaction costs. Firms with financial hedging are found to experience lower external financing costs (e.g., Campello *et al.*, 2011; Chen and King,

2014). Finally, because of the relative consistency of a firm's hedging policy, an acquirer that hedges its financial risk exposures before the deal announcement is expected to continue to manage its financial risk exposures after deal completion. The expectation of the future risk reduction of the combined firm will generate a positive stock market reaction at the deal announcement.⁸

H2: *Cross-border M&As carried out by derivatives users have a higher probability of completion than those carried out by non-users.*

The probability of deal completion will increase with an acquirer's hedging experience for four reasons. First, because it is costly to establish hedging programmes and the hedging positions will result in a contractual commitment to carry out the deal, a pre-established FX hedging position in the target nation's currency signals the acquirer's commitment to the deal. Second, an established financial hedging programme can help an acquirer manage deal-related financial risk and reduce transaction risk exposures. It is less likely for an acquirer with a hedging programme to withdraw a deal due to unexpected and unfavourable market conditions in the exchange rate or the IR. Third, hedging mitigates information asymmetry between investors and managers, which in turn reduces an acquirer's stock return volatility. For deals involving stock payments, target shareholders are thus more willing to accept acquirer stocks. Finally, financial hedging can lower an acquirer's external financing costs, so that a derivatives user can offer more competitive deal terms to the target.

H3: *Financial derivatives users experience longer deal completion time on cross-border M&As than non-users.*

Offenberg and Pirinsky (2015) argue that the M&A deal completion time is the result of a trade-off between the costs and benefits of waiting. The use of FCDs and IRDs can reduce an acquirer's financial risk exposure during the interim period, leading to lower waiting costs. Derivatives users can thus afford to take longer to review the transaction details and negotiate more favourable deal terms than non-users can. In other words, because of the reduced FX risk and IR risk in the interim period, an acquirer with financial hedging programmes has less pressure to rush through the deal and can take more time to appraise the deal terms. Additionally, established hedging programmes demonstrate the conservative risk attitude of an acquirer that is inclined to take more time to carefully assess the transaction details and potential deal risks.

H4: *Financial derivatives users have better post-deal long-run performance than non-users after cross-border M&As.*

An acquirer attains increased foreign operations after cross-border deal completion. This changes the acquirer's FX risk exposures in the long term. As a result, the benefits of hedging could extend well beyond the deal announcement. Derivatives users could have better long-run performance than non-users for the following three reasons. First, related to Hypothesis 3, the more deliberate deal consideration by derivatives users can lead to

⁸ We acknowledge that, in some cases, an acquirer purchases a foreign target as an operational hedge for its existing FX risk exposure. This will reduce the benefits of financial hedging and alleviate the positive effect of hedging on acquirer announcement returns.

higher-quality targets and more favourable deal terms. Second, certain types of derivatives put in place before the deal announcement, such as swaps, can continue to reduce the combined firm's risk exposure after deal completion. Finally, an acquirer that manages its risk exposures before the deal announcement is expected to continue to manage its exposure post-deal.

4. Data

4.1. *Sample selection and variable construction*

The data used in this paper are from several sources. We start with all cross-border M&As with announcement dates between 2000 and 2014 from the Thomson Reuters Securities Data Company (SDC) Platinum Mergers and Acquisitions database.⁹ We require the acquirer to be publicly traded in the United States and the target to be a foreign company. Applying the standard filters used in the literature, we then exclude all transactions that are labelled as a minority stake purchase, acquisition of remaining interest, privatisation, repurchase, exchange offer, self-tender, recapitalisation, or spinoff. We further require a minimum deal value of US\$ 1 million and a minimum acquirer market value of US\$ 20 million. In line with previous M&A studies, the percentage of target shares held by the acquirer is restricted to less than 10% prior to the transaction and more than 50% after the transaction. Following Allayannis and Weston (2001) and Bartram *et al.* (2011), deals with acquirers from the financial or utilities industry are dropped. This procedure yields an initial sample of 2,753 observations. We further set the restriction that acquirers be included in the S&P 1500 list when the deals were announced.¹⁰ Finally, after merging these data with the Center for Research in Securities Prices (CRSP) and Compustat data, we obtain a sample with 1,385 cross-border M&A deals.

For each of the 1,385 M&As, we manually collect the acquirer's financial hedging data from the acquirer's 10-K or 10-K405 reports filed in the fiscal year prior to the deal announcement. All reports are retrieved from the Securities and Exchange Commission (SEC) Electronic Data Gathering, Analysis and Retrieval (EDGAR) electronic filing system. We require that an acquirer have filed at least one 10-K or 10-K405 report when the deal was announced. Our final sample consists of 1,369 deals.¹¹

⁹ Our sample period begins in 2000 because the new US Financial Accounting Standard Board (FASB) standard for disclosure of derivatives transactions, Statement of Financial Accounting Standard No. 133 (SFAS 133, Accounting for Derivative Instruments and Hedging Activities) became effective in June 1999. For detailed information, see www.fasb.org/summary/stsum133.shtml.

¹⁰ The S&P 1500, or S&P Composite 1500 index, combines all stocks in three leading indices: the S&P 500, the S&P Mid Cap 400, and the S&P Small Cap 600. It covers approximately 90% of the US stock market capitalisation. We restrict our sample to within the S&P 1500 because certain control variables, such as corporate governance, are only available for these companies.

¹¹ Following the international finance and cross-border M&A literature, we delete observations in which the acquirer nations are Bermuda, the Cayman Islands, Ecuador, and the Netherlands Antilles.

We focus on the use of FCDs and IRDs because they are directly related to the management of cross-border M&A risk exposures. Information on CDs is also collected for our robustness tests. Following the corporate financial hedging literature,¹² we create the following hedging variables: 1) *Fcd*, a 0/1 binary variable indicating whether a firm hedges FX risk; 2) *Fcd_target*, a 0/1 binary variable indicating whether a firm hedges FX risk between the US dollar and the target nation's currency; 3) *Ird*, a 0/1 binary variable indicating whether a firm hedges IR risk; 4) *Fcd/Ird*, a 0/1 binary variable that equals one if a firm hedges either FX risk or IR risk and zero if the firm hedges neither; 5) *Hedging_scope*, a 0/1/2 categorical variable that equals two if a firm hedges both FX and IR risk, one if a firm hedges only one of these two risks, and zero if a firm hedges neither; 6) *Nv_derivatives*, the aggregate notional value of FCDs and IRDs, normalised by the acquirer's total assets; and 7) *Commodity*, a 0/1 binary variable indicating whether a firm hedges commodity price risk. We also collect the foreign currency-denominated debt information and create a binary indicator variable, *Foreign_debt*, and use it as a control variable in our empirical analysis.¹³

We obtain firm accounting data from Compustat, foreign sales information from Compustat Segments files, stock return and factor return data from the CRSP, financial analyst data from the Institutional Brokers Estimate System (IBES), and corporate governance information from the Institutional Shareholder Services (ISS, formerly RiskMetrics). Of the 1,369 sample deals, we are able to collect geographical segment information for 1,323 of them. Following Allayannis and Weston (2001), we assume that, for the remaining 46 deals, the acquirers have no foreign sales. In our robustness tests, we obtain the managerial ability scores (Demerjian *et al.*, 2012) from Peter Demerjian's website¹⁴ and count the mentions of target nations in acquirers' 10-K reports. A detailed description of variables and data sources can be found in the Appendix.

4.2. Descriptive statistics

Figure 2 shows the distribution of our cross-border M&A sample by announcement year over the sample period 2000–2014. Consistent with Harford (2005), we find a merger wave pattern in our sample. The total number of deals drops twice following the burst of the dot-com bubble in 2000 and the global financial crisis around 2008. The number of

¹² We search the following keywords in acquirer annual reports to locate the information of financial derivatives: *Item 7A*, *derivative*, *derivative(s)*, *instrument(s)*, *hedge*, *financial instrument*, *swap*, *futures*, *forward contract*, *forward exchange*, *option contract*, *risk management*, *foreign currency*, *currency exchange*, *notional*, *fair value*, *borrowing*, *debt*, *credit facilities*, *line(s) of credit*, *notes payable*, *commodity* and *commodities*. This list of keywords is compiled based on those used in previous corporate financial hedging studies (e.g., Allayannis and Ofek, 2001; Allayannis and Weston, 2001; Graham and Rogers, 2002; Bartram *et al.*, 2011; Campello *et al.*, 2011). When a keyword is found, we read the surrounding text and hand-code our hedging variables.

¹³ We supplement foreign currency debt information using bond data from SDC's Global New Issues dataset.

¹⁴ <http://faculty.washington.edu/pdemerj>.

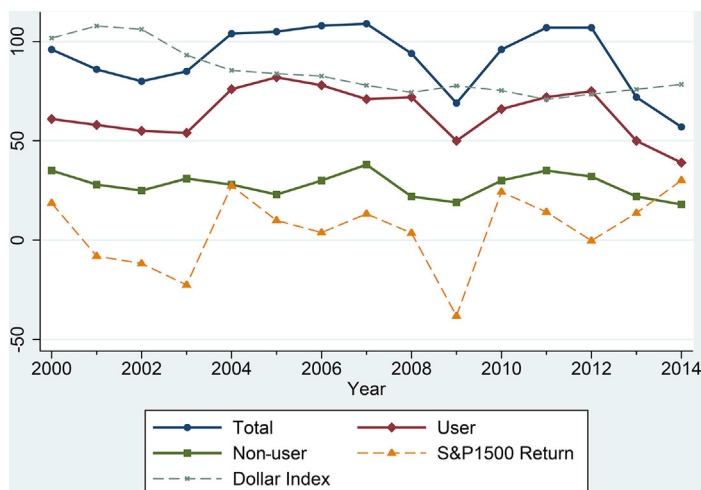


Fig. 2. Distribution of cross-border M&As by year

This figure presents the annual numbers of cross-border M&As deals made by S&P 1500 companies between 2000 and 2014. We also plot the annual numbers of cross-border M&As made by derivatives users and non-users using two solid lines. The two dotted lines represent the S&P 1500 index annual returns (multiplied by 100) and the annual trade-weighted US dollar index levels. [Color figure can be viewed at wileyonlinelibrary.com.]

deals involving derivatives users and non-users exhibits a very similar time-series pattern as the whole sample. Figure 2 also shows the S&P 1500 index annual return (multiply by 100) and the trade-weighted US dollar index. The annual deal number exhibits a positive correlation with the S&P 1500 index return. This confirms the findings documented in previous studies that merger activities are positively correlated with the valuation of the stock market (e.g., Shleifer and Vishny, 2003; Rhodes-Kropf and Viswanathan, 2004; Rhodes-Kropf *et al.*, 2005). It can also be observed that the trade-weighted US dollar index exhibits high volatility during our sample period. The volatile exchange rate provides an incentive for acquirers to use derivatives to reduce cross-border deal-related FX risk exposures.

Panel A of Table 1 presents the distribution of our cross-border M&A sample by target nation/region. Our sample includes 1,369 deals with non-US targets from a total of 58 different nations/regions. The top five target nations are the United Kingdom (294), Canada (219), Germany (134), France (89), and Australia (57). Panel B of Table 1 presents the industry distribution of acquirers in our sample according to the Fama–French 10 industry classifications (Fama and French, 1997). Business equipment, manufacturing, and healthcare are the top three industries ranked by the number of M&A deals, accounting for a total of 74.6% of our sample observations. Other; oil, gas, and coal; and consumer durables are the three industry groups with the smallest number of observations and together they account for 13% of total deals. Panels A and B demonstrate that our sample includes targets from a wide range of nations/regions and acquirers from diversified industries.

Panel C of Table 1 presents the summary statistics of our main hedging variables. Among the 1,369 M&As, 63.2% of acquirers use FCDs, 43.2% of acquirers use IRDs,

Table 1
Descriptive statistics

Panel A. Distribution of cross-border M&As by target country/region. This panel presents the numbers of cross-border M&A deals by target nation/region. Our final sample includes 1,369 cross-border M&As between 2000 and 2014. All acquirers are S&P 1500 companies in the United States and the targets are from 58 different foreign countries or regions.

Nation	Freq.	Pct.	Nation	Freq.	Pct.	Nation	Freq.	Pct.
United Kingdom	294	21.48	Sweden	37	2.70	Japan	15	1.10
Canada	219	16.00	India	36	2.63	Mexico	14	1.02
Germany	134	9.79	Brazil	32	2.34	Spain	13	0.95
France	89	6.50	Norway	22	1.61	Austria	5	0.37
Australia	57	4.16	Belgium	21	1.53	Luxembourg	5	0.37
Israel	48	3.51	Denmark	20	1.46	New Zealand	5	0.37
Switzerland	46	3.36	Italy	20	1.46	South Africa	5	0.37
Netherlands	44	3.21	South Korea	19	1.39	Egypt	4	0.29
China	37	2.70	Ireland-Rep	18	1.31	Others	37	2.69
						Total	1,369	100

Panel B. Distribution of cross-border M&As by acquirer industry. This panel presents the numbers of cross-border M&A deals by acquirer industry. Our sample includes 1,369 cross-border M&As between 2000 and 2014. All acquirers are S&P 1500 companies and the targets are from 58 different foreign countries or regions. We assign 1,369 deal acquirers into Fama–French 10 industries based on acquirer SIC codes. Financial and public utilities industries are not included in our sample.

Fama–French 10 industries	Frequency	Percent
Business Equipment – Computers, Software, and Electronic Equipment	485	35.43
Manufacturing – Machinery, Trucks, Planes, Chemicals, Off Furn., Paper, Com. Printing	380	27.76
Healthcare, Medical Equipment, Drugs	156	11.40
Consumer Nondurables – Food, Tobacco, Textiles, Apparel, Leather, Toys	89	6.50
Wholesale, Retail, and Some Services (Laundries, Repair Shops)	81	5.92
Other – Mines, Constr., BldMt, Trans, Hotels, Bus Serv, Entertainment	78	5.70
Oil, Gas, and Coal Extraction and Products	59	4.31
Consumer Durables – Cars, TVs, Furniture, Household Appliances	41	2.99
Total	1,369	100

Panel C. Summary statistics of hand-coded variables. This panel presents the 1,369 cross-border M&A acquirers' hedging information which we collect from their 10-K reports on EDGAR. The sample period is between 2000 and 2014. All acquirers are S&P 1500 companies and the targets are from 58 different foreign countries or regions. All the variables are constructed in the fiscal year before the deal announcement. Detailed definitions of all variables can be found in the Appendix.

Variable	Obs.	Mean	Std. Dev.	p25	p50	p75
<i>Fcd</i>	1,369	0.632	0.482			
<i>Fcd_target</i>	1,219	0.374	0.484			
<i>Ird</i>	1,369	0.432	0.495			
<i>Fcd/Ird</i>	1,369	0.726	0.446			
<i>Hedging_scope</i>	1,369	1.064	0.780	1	1	2
<i>Nv_derivatives</i> (for hedgers)	820	0.129	0.371	0.025	0.078	0.159
<i>Foreign_debt</i>	1,369	0.199	0.399			
<i>Commodity</i>	1,369	0.198	0.399			

Panel D. Summary statistics of the key dependent variables. This panel presents summary statistics of the key dependent variables. The sample includes 1,369 cross-border M&A deals between 2000 and 2014. All acquirers are S&P 1500 companies and the targets are from 58 different foreign countries or regions. Summary statistics are presented for the full sample, the financial derivatives users sample, and the nonusers sample. Financial derivatives users are defined by *Fcd/Ird*. The last column (*Diff*) reports the significance levels of the *t*-test of the difference between the two sub-groups. Detailed definitions of all the variables can be found in the Appendix. Significance at the 0.01, 0.05 and 0.10 levels is indicated by ***, ** and *, respectively.

Variable	Full Sample			Derivatives User			Non-user			Diff
	Obs.	Mean	Std. Dev.	Obs.	Mean	Std. Dev.	Obs.	Mean	Std. Dev.	
<i>CAR_[-5,5]</i>	1,369	0.47%	7.67%	994	0.62%	9.22%	375	-0.58%	7.05%	***
<i>Completion</i>	1,369	0.932	0.252	994	0.927	0.261	375	0.947	0.225	
<i>Completion_time</i>	1,276	48.8	85.6	921	51.2	71.7	355	42.6	114.1	*
$\Delta ROA_{t,t+3}$	945	-0.07%	6.63%	695	0.17%	5.72%	250	-0.76%	8.65%	*
<i>BHAR_3Y</i>	1,040	-0.01%	2.36%	747	0.11%	2.19%	293	-0.32%	2.73%	***

Panel E. Summary statistics of control variables. This panel presents summary statistics for 1,369 cross-border M&A deals in our sample. The sample period is between 2000 and 2014. All acquirers are S&P 1500 companies and the targets are from 58 different foreign countries or regions. Summary statistics are presented for the full sample, the financial derivatives (*Fcd/Ird*) users sample, and the non-users sample. The last column reports the significance levels of the *t*-test on the difference between the two sub-groups. Detailed definitions of all the variables can be found in the Appendix. Significance at the 0.01, 0.05 and 0.10 levels is indicated by * *, * and *, respectively.

Variable	Full Sample			Derivatives User			Non-user		
	Obs.	Mean	Std. Dev.	Obs.	Mean	Std. Dev.	Obs.	Mean	Std. Dev.
Deal characteristics									
<i>Cash</i>	1,369	0.438	0.496	994	0.438	0.496	375	0.437	0.497
<i>Equity</i>	1,369	0.064	0.245	994	0.048	0.214	375	0.107	0.309
<i>Non-public</i>	1,369	0.869	0.337	994	0.867	0.340	375	0.875	0.332
<i>Toehold</i>	1,369	0.006	0.076	994	0.007	0.084	375	0.003	0.052
<i>Hostile</i>	1,369	0.005	0.071	994	0.005	0.071	375	0.005	0.073
<i>Tender</i>	1,369	0.061	0.240	994	0.065	0.247	375	0.051	0.220
<i>Related_industry</i>	1,369	0.573	0.495	994	0.561	0.496	375	0.605	0.489
<i>Relative_size</i>	1,368	0.071	0.151	993	0.066	0.143	375	0.084	0.168
<i>Transaction_value</i>	1,368	491.2	3,686	993	594.8	4,290.7	375	216.9	847.1
Acquirer characteristics									
<i>Size</i>	1,369	8.49	1.69	994	8.82	0.05	375	7.62	0.07
<i>Tobin's Q</i>	1,369	2.280	1.696	994	2.243	1.689	375	2.376	1.716
<i>Leverage</i>	1,369	0.434	0.193	994	0.471	0.185	375	0.336	0.180
<i>Cash/assets</i>	1,368	0.168	0.164	993	0.148	0.146	375	0.222	0.193
<i>Governance</i>	1,321	1.888	1.269	973	1.859	1.292	348	1.968	1.199
<i>Runup</i>	1,369	0.092	0.338	994	0.083	0.296	375	0.117	0.429
<i>Sigma</i>	1,369	0.020	0.009	994	0.018	0.009	375	0.024	0.011
<i>ROA</i>	1,369	0.153	0.077	994	0.155	0.071	375	0.147	0.092
<i>Managerial_score</i>	1,220	0.026	0.143	883	0.023	0.143	337	0.036	0.143
<i>Counts</i>	1,369	23.20	78.55	994	22.29	71.72	375	25.62	94.37

and 72.6% acquirers hold at least one of these two types of derivatives. The proportion of derivatives users in our sample is higher than reported by Bartram *et al.* (2011) (FCD 37.8% and IRD 40.4%) or Campello *et al.* (2011) (FCD 27.3% and IRD 35.6%).¹⁵

Since SFAS 133 became effective in 2000, it is no longer mandatory for US public firms to report the notional value of their derivatives contracts, as previously required by SFAS 119. However, among 994 acquirers with hedging programmes in our sample, 820 still voluntarily report the notional value of their derivatives contracts. On average, the total notional value of FCDs and IRDs accounts for 12.9% of the total assets of these acquirers. Panel C also reports that 19.9% of acquirers have outstanding foreign currency-denominated debt and 19.8% of acquirers use CDs to hedge commodity price risks.¹⁶

Panel D of Table 1 reports the summary statistics of our key dependent variables of interest: acquirer announcement returns ($CAR_{[-5, 5]}$), the deal completion probability (*Completion*), the number of days between the M&A announcement and deal completion (*Completion_time*), acquirer long-run operating performance ($\Delta AROA_{t, t+3}$), and acquirer long-run stock performance ($BHAR_{3Y}$). Summary statistics for the full, derivatives users, and non-users samples are reported. On average, financial derivatives users have significantly better announcement returns, longer deal completion time, and better long-run performance than non-users. The difference in the deal completion probability between the user and non-user samples is not statistically significant.

Panel E of Table 1 presents the summary statistics of deal and acquirer characteristics. We report the number of observations and the mean and standard deviation of each variable for the full, derivatives users, and non-users sample, respectively. The last column presents the statistical significance of the mean difference tests between the derivatives users and non-users samples. In total, we have 1,276 completed deals and 93 withdrawn deals. Deals made by derivatives users have a higher absolute transaction value but a lower relative deal size compared with acquirer size. Deals made by derivatives users are also less likely to involve equity as a payment method. Hedgers are more likely to have larger firm size, higher leverage, and greater pre-deal ROA. On the

¹⁵ Allayannis and Weston (2001) find that the number of public firms using derivatives to manage financial risks increases over time. Our sample period is 2000–2014, which is later than the sample periods of Bartram *et al.* (2011) (2000–2001) and Campello *et al.* (2011) (1996–2002). In addition, previous hedging studies have shown a positive relationship between firm size and the use of derivatives. Our sample firms are much larger than those in the previous studies because we focus on S&P 1500 firms capable of carrying out cross-border M&As, while Bartram *et al.* (2011) and Campello *et al.* (2011) do not have such sample restrictions.

¹⁶ In a survey of 392 chief financial officers by Graham and Harvey (2001), 85.8% of the participants chose ‘providing a “natural hedge”’ as an important factor in their decision about issuing foreign debt. In a recent study, Clark and Judge (2016) compare the effect of different FX hedging strategies on firm value and conclude that financial derivatives are used by firms to hedge short-term risk exposures while foreign currency-denominated debts are used to hedge long-term risk exposures. Previous hedging studies have also examined the use of CDs for firms in certain industries with strong commodity price risk exposure (e.g., Carter *et al.*, 2006; Jin and Jorion, 2006). Commodity price risk is not directly related to cross-border M&As and only affects a small number of industries; so hedging by CDs is expected to be much less correlated with deal outcomes than hedging by FCDs or by IRDs.

other hand, hedgers are found to have smaller cash holdings, runups, and sigma. We do not find a statistically significant difference in Tobin's Q between the user and non-user groups.¹⁷

5. Empirical Results

5.1. Financial hedging and acquirer CARs: Ordinary least squares (OLS)

To examine the effect of financial hedging on cross-border deal performance, we study acquirer CARs estimated by a market model with the CRSP value-weighted index. Following Golubov *et al.* (2015), the market model is estimated using at least 30 non-missing daily return data over the (−300, −91) period prior to the deal announcement. Acquirer CARs are measured over a window of (−5, +5), where day 0 is the deal announcement date.¹⁸

Table 2 reports the results of OLS regressions with robust standard errors for the 1,276 completed deals. The dependent variables in all seven regressions are the acquirer announcement CARs. The independent variables of interest are the corporate financial hedging proxy variables: the FCD user indicator (*Fcd*), the derivatives user indicator (*Fcd/Ird*),¹⁹ the hedging scope indicator (*Hedging_scope*), and the notional value of financial derivatives normalized by acquirers' total assets (*NV_derivatives*). We control for year, Fama–French 10 industry, and S&P index fixed effects in all regressions.

In columns (1) to (4) of Table 2, *Fcd* is the key independent variable of interest. In column (1), we do not include any control variables. We control for deal characteristics in column (2) and then add acquirer characteristics in column (3). In column (4), we add one more control variable, *Foreign_debt*, to control for the impact of foreign currency-denominated debt on acquirer FX exposures (Allayannis *et al.*, 2003; Kedia and Mozumdar, 2003; Clark and Judge, 2016). The coefficients of *Fcd* remain positive and statistically significant in all four regressions. These results show that acquirers engaged in FX risk hedging activities have significantly higher announcement CARs than acquirers that do not. The improvement of CARs is not only statistically significant but also economically important. Using the regression model specified in column (4), we estimate that FCD users experience an average of 1.0% higher announcement CARs than non-users. This is equivalent to an increase of US\$ 193.7 million in shareholder value for an average-sized acquirer.

¹⁷ This result does not conflict with the findings in the previous hedging literature, because Tobin's Q is not compared cross-sectionally in our sample.

¹⁸ As a robustness check, we also estimate the market model over the alternative period of (−260, −60), (−200, −20) and (−200, −60) and check alternative CAR estimation windows of (−3, +5) and (−1, +5). In addition, we estimate acquirer CARs using the Fama–French three-factor model (Fama and French, 1993) and the Carhart (1997) four-factor model. The results are qualitatively the same using these specifications.

¹⁹ We do not include *Ird* as an independent hedging variable because the direct effect of IR risk hedging on acquirer CARs is conditional on the issuance of debt by acquirers to finance cross-border M&As. Instead, we include *Fcd/Ird* as a hedging variable to account for whether an acquirer has an established hedging programme or not.

Table 2

Acquirer announcement returns estimated by the market model and financial hedging

This table presents the regression results of acquirer CARs on financial hedging variables. The OLS regressions with robust standard errors are based on a sample of 1,274 completed deals that are carried out by S&P 1500 firms between 2000 and 2014. The dependent variable is the acquirer CAR over the 11-day event window $(-5, +5)$, where day 0 is the announcement day. The benchmark is estimated by the market model with the CRSP value-weighted index over the pre-announcement window $(-300, -91)$. Detailed definitions of all the variables can be found in the Appendix. Year, Fama–French 10 industry, and S&P index fixed effects are controlled for all regressions. The p -values are reported in parentheses. Significance at the 0.01, 0.05 and 0.10 levels is indicated by ***, ** and *, respectively.

	1	2	3	4	5	6	7
<i>Fcd</i>	0.012** (0.016)	0.011** (0.032)	0.010* (0.055)	0.010* (0.056)			
<i>Fcd/Ird</i>					0.011* (0.055)		
<i>Hedging_scope</i>						0.005* (0.098)	
<i>Nv_derivatives</i>							0.015** (0.036)
<i>Cash</i>		-0.000 (0.984)	-0.001 (0.844)	-0.001 (0.850)	-0.001 (0.815)	-0.001 (0.825)	-0.001 (0.847)
<i>Equity</i>		-0.033** (0.021)	-0.031** (0.038)	-0.031** (0.038)	-0.031** (0.037)	-0.031** (0.036)	-0.031*** (0.004)
<i>Non-public</i>		-0.014 (0.104)	-0.007 (0.407)	-0.007 (0.409)	-0.008 (0.388)	-0.007 (0.404)	-0.009 (0.338)
<i>Toehold</i>		0.029 (0.203)	0.038* (0.083)	0.038* (0.081)	0.038* (0.088)	0.037* (0.087)	0.031 (0.323)
<i>Hostile</i>		-0.013 (0.351)	-0.020 (0.138)	-0.019 (0.154)	-0.020 (0.150)	-0.020 (0.133)	-0.022 (0.563)
<i>Tender</i>		-0.011 (0.341)	-0.015 (0.225)	-0.015 (0.225)	-0.015 (0.205)	-0.015 (0.213)	-0.014 (0.285)
<i>Related_industry</i>		-0.002 (0.684)	-0.004 (0.367)	-0.004 (0.362)	-0.004 (0.322)	-0.004 (0.310)	-0.006 (0.188)
<i>Relative_size</i>			0.042* (0.063)	0.042* (0.062)	0.041* (0.064)	0.041* (0.062)	0.027 (0.114)
<i>Size</i>			-0.002 (0.309)	-0.002 (0.296)	-0.002 (0.351)	-0.003 (0.256)	-0.001 (0.617)
<i>Tobin's Q</i>			-0.001 (0.689)	-0.001 (0.701)	-0.001 (0.692)	-0.001 (0.782)	-0.002 (0.348)
<i>Leverage</i>			-0.005 (0.698)	-0.005 (0.677)	-0.006 (0.651)	-0.006 (0.657)	-0.009 (0.542)
<i>Cash/assets</i>			-0.015 (0.433)	-0.015 (0.455)	-0.013 (0.516)	-0.013 (0.493)	-0.016 (0.397)
<i>Governance</i>			-0.001 (0.596)	-0.001 (0.604)	-0.001 (0.707)	-0.001 (0.657)	-0.002 (0.505)
<i>Runup</i>			-0.017* (0.067)	-0.018* (0.065)	-0.018* (0.063)	-0.018* (0.060)	-0.018** (0.016)

Table 2
Continued

	1	2	3	4	5	6	7
<i>Sigma</i>			−0.014 (0.981)	−0.008 (0.990)	0.051 (0.930)	0.031 (0.958)	0.128 (0.758)
<i>Foreign_debt</i>				0.002 (0.694)	0.002 (0.681)	0.002 (0.733)	0.003 (0.672)
<i>Intercept</i>	−0.007 (0.543)	0.013 (0.415)	0.025 (0.375)	0.026 (0.370)	0.021 (0.471)	0.028 (0.340)	0.015 (0.625)
S&P index fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,274	1,274	1,228	1,228	1,228	1,228	1,075
Adj R-squared	0.044	0.056	0.055	0.065	0.065	0.064	0.072

We then replace the hedging variable of *Fcd* with *Fcd/Ird*, which is a broader indicator of hedging that takes into account the use of both FCDs and IRDs. The coefficient of *Fcd/Ird* is also positive and statistically significant, as reported in column (5) of Table 2. The hedging effect remains statistically significant when we use *Hedging_scope* as the hedging variable in column (6) and *NV_derivatives*²⁰ in column (7). Overall, our results are consistent with Hypothesis 1, that derivatives users have higher acquirer CARs than non-users. In addition, we find acquirers that hedge more types of risks (*Hedging_scope*) and have higher derivatives contract values (*NV_derivatives*) experience higher announcement returns.

5.2. Financial hedging and acquirer CARs: The Heckman two-stage procedure

Endogeneity concerns about our test results emerge if financial hedging decisions are not randomly distributed among firms but are, instead, related to certain unobservable firm characteristics that have a positive impact on acquirer announcement returns. If that is the case, an acquirer that uses financial derivatives would have higher CARs, whether it engages in financial hedging or not. The ideal approach to address this self-selection bias is to compare the deal outcomes of the same acquirer with and without financial hedging.

²⁰ Over our sample period 2000–2014, US public companies were only required to report the fair value of their derivatives positions but not the notional value. Although we find that most companies that report derivative information in their annual report choose to report both values, the notional value of derivatives as measured by *NV_derivatives* is still a noisy proxy for hedging and suffers from self-selection bias in reporting. Therefore, we focus on hedging indicator variables in the rest of our empirical analysis.

Since this is not practical, we adopt the Heckman treatment effect model: a linear regression model augmented with an endogenous binary treatment variable (Heckman, 1978; Wooldridge, 2010). The estimation is conducted by a two-step consistent estimator. The first-step treatment equation models the choice of using financial derivatives and the second-step outcome equation corrects for the omitted variables and selection bias.

The key to implementing the two-step treatment effect estimation is to identify instrument variables that strongly correlate to a firm's hedging decision but not its announcement return. We employ two instrument variables in our tests: *Analyst_number* and *Foreign_sales/Sales*. Geczy *et al.* (1997) find that both the number of financial analyst firms following and the ratio of foreign sales to total sales are related to the likelihood that a firm uses currency derivatives. The number of financial analysts following a firm could measure information asymmetry so that more analysts following are positively related to the availability of information, leading to less of a need for managers to use derivatives hedging as a signalling tool (DeMarzo and Duffie, 1995). An alternative explanation is that the number of analysts following is a proxy for the pressure that managers receive for their firm performance (Geczy *et al.*, 1997). Thus, the more analysts following a firm, the higher the probability that the firm will engage in financial hedging to reduce variation in firm performance. The ratio of foreign sales to total sales measures an acquirer's exposure to foreign currency risk before the M&A deal. Therefore, higher foreign currency risk exposure is associated with greater potential hedging benefits and leads to a higher probability of FX hedging (Kedia and Mozumdar, 2003).²¹

Table 3 presents the two-step treatment effect test results. In the first-step treatment regressions, we conduct probit tests in which the dependent variable of interest is *Fcd* in models 1–2 and *Fcd/Ird* in models 3–4. The variable *Analyst_number* is the instrument variable employed in models 1 and 3, while *Foreign_sales/Sales* is the instrument variable employed in models 2 and 4. The coefficients of both instrument variables are statistically significant in the first-step treatment equations. In the second-step outcome equations, the dependent variables are acquirer CARs estimated by the market model. Similar to the OLS regression results, the coefficients of *Fcd* and *Fcd/Ird* are both positive and statistically significant in the second-step regressions. The Heckman two-stage test results confirm our findings of a positive relationship between hedging and acquirers CARs when endogeneity is addressed.

²¹ The pairwise correlation between acquirer CARs and the number of financial analysts following is -0.09 and the pairwise correlation between acquirer CARs and the ratio of foreign sales to total sales is 0.03 . To avoid overidentification, we use *Analyst_number* and *Foreign_sales/Sales* separately as instrument variables in the first-step treatment equation. Our results are qualitatively the same when we combine the two instrument variables together in the first-step treatment equation. Unlike standard instrument variables and two-stage least squares regressions, the Heckman model does not require exclusion restrictions to be estimated (e.g., Li and Prabhala, 2007; Golubov *et al.*, 2012). The relevance condition of our two instruments is examined in the first-stage regression. The corresponding p -value for the joint test of whether the excluded instruments are significant is 0.000 , suggesting that the relevance condition is satisfied.

Table 3
Acquirer CARs and financial hedging: The treatment effect

This table presents the regression results of acquirer CARs on financial hedging variables, using an endogenous binary treatment model estimated by a two-step consistent estimator. The first-step treatment equation is estimated by a probit regression, where the dependent variable is *Fcd* in model 1–2 and *Fcd/lrd* in model 3–4. In models 1 and 3, the instrument variable used in the treatment equation is *Analyst_number*. In models 2 and 4, the instrument variable used in the treatment equation is *Foreign_sales/Sales*. In the outcome equations, the dependent variable is acquirer CARs estimated by the market model. Detailed definitions of all the variables can be found in the Appendix. Year, industry, and S&P index fixed effects are controlled for in all regressions. The *p*-values are reported in parentheses. Significance at the 0.01, 0.05 and 0.10 levels is indicated by ***, ** and *, respectively.

	Model 1		Model 2		Model 3		Model 4	
	Treatment	Outcome	Treatment	Outcome	Treatment	Outcome	Treatment	Outcome
<i>Fcd</i>		0.109*** (0.009)		0.034* (0.062)				
<i>Fcd/lrd</i>						0.104*** (0.001)		0.066*** (0.003)
<i>Cash</i>		−0.002 (0.698)		−0.001 (0.872)		−0.003 (0.584)		−0.001 (0.809)
<i>Equity</i>		−0.029*** (0.005)		−0.029*** (0.005)		−0.030*** (0.003)		−0.029*** (0.005)
<i>Non-public</i>		−0.010 (0.263)		−0.005 (0.551)		−0.011 (0.242)		−0.006 (0.510)
<i>Toehold</i>		0.051 (0.138)		0.056* (0.063)		0.054 (0.111)		0.057* (0.063)
<i>Hostile</i>		−0.027 (0.488)		−0.024 (0.527)		−0.027 (0.482)		−0.022 (0.556)
<i>Tender</i>		−0.009 (0.467)		−0.014 (0.246)		−0.010 (0.440)		−0.014 (0.232)
<i>Related_industry</i>		−0.003 (0.527)		−0.004 (0.308)		−0.003 (0.449)		−0.004 (0.307)
<i>Relative_size</i>		0.037**		0.047***		0.036**		0.046***

Table 3
Continued

	Model 1		Model 2		Model 3		Model 4	
	Treatment	Outcome	Treatment	Outcome	Treatment	Outcome	Treatment	Outcome
<i>Size</i>	0.366*** (0.000)	(0.021) -0.011** (0.014)	0.284*** (0.000)	(0.003) -0.004 (0.167)	0.265*** (0.000)	(0.026) -0.007** (0.036)	0.197*** (0.002)	(0.004) -0.004 (0.158)
<i>Tobin's Q</i>	-0.094** (0.016)	0.002 (0.437)	-0.070* (0.071)	0.000 (0.892)	-0.081** (0.046)	0.001 (0.552)	-0.059 (0.147)	0.000 (0.842)
<i>Leverage</i>	1.245*** (0.000)	-0.050** (0.037)	1.630*** (0.000)	-0.013** (0.406)	1.570*** (0.000)	-0.047** (0.022)	1.896*** (0.000)	-0.027 (0.124)
<i>Cash/assets</i>	0.175 (0.622)	-0.011 (0.623)	-0.097 (0.787)	-0.014 (0.449)	-0.304 (0.414)	0.008 (0.708)	-0.440 (0.242)	-0.003 (0.879)
<i>Governance</i>	-0.025 (0.565)	-0.002 (0.464)	0.036 (0.410)	-0.001 (0.719)	-0.113** (0.017)	0.000 (0.855)	-0.085* (0.063)	0.001 (0.662)
<i>Runup</i>	-0.227 (0.129)	-0.008 (0.415)	-0.066 (0.656)	-0.018** (0.015)	-0.191 (0.219)	-0.009 (0.284)	-0.049 (0.746)	-0.018** (0.021)
<i>Sigma</i>	7.637 (0.325)	-0.275 (0.562)	4.782 (0.531)	-0.216 (0.589)	-9.041 (0.257)	0.237 (0.614)	-11.013 (0.159)	0.069 (0.868)
<i>Foreign_debt</i>	0.137 (0.255)	-0.002 (0.767)	-0.017 (0.886)	0.001 (0.907)	0.064 (0.632)	0.001 (0.884)	0.046 (0.725)	0.000 (0.932)
<i>Analyst_number</i>	-0.021** (0.029)				-0.015* (0.095)			
<i>Foreign_sales/Sales</i>			1.672*** (0.000)				1.068*** (0.000)	
<i>Intercept</i>	-3.650*** (0.000)	0.079** (0.037)	-3.780*** (0.000)	0.033 (0.276)	-2.109*** (0.001)	0.020 (0.560)	-2.149*** (0.000)	0.009 (0.771)
S&P/Industry/Year fixed effects	YES	YES	YES	YES	YES	YES	YES	YES
Observations	1,081	1,081	1,184	1,184	1,081	1,081	1,184	1,184
<i>Prob > chi2</i>	0.000		0.000		0.000		0.000	

5.3. Other deal-related outcomes

In this section, we investigate whether financial hedging affects deal completion probability and deal completion time.

5.3.1. Financial hedging and deal completion probability. Columns (1) to (4) of Table 4 report the results of the probit regressions of deal completion probabilities on three measures of acquirer financial hedging activities: *Fcd*, *Fcd/Ird*, and *Fcd_target*. In the first two regressions, the coefficients of the financial hedging variables, *Fcd* and *Fcd/Ird*, are not statistically significant. In the remaining regressions, the hedging variable employed is *Fcd_target*, which is a dichotomous variable indicating if the acquirer uses currency derivatives specifically to hedge the target nation's currency risk. We find the regression coefficients for *Fcd_target* to be positive and statistically significant in the probit regressions reported in columns (3) and (4). This suggests that, if an acquirer holds the target nation's currency derivatives contracts prior to the deal announcement, the probability of deal completion is higher. Using the model specification of column (4) as an example, the untabulated marginal effect coefficients show that, with all the other independent variables at their mean value, acquirers that hedge the target country's currency risk have a 4% higher probability of completing their deals than acquirers that do not. As a robustness check, we then conduct a logit regression with the same control variables as in column (4). The results are reported in column (5), which show that the coefficient of *Fcd_target* remains positive and statistically significant. Overall, our findings are consistent with Hypothesis 2.

Apart from the impact of the hedging policy, we find that a deal is more likely to be successful when the target is a non-public rather than a public firm. One possible explanation is that shares of private companies are more concentrated (Golubov *et al.*, 2012) and the voting process for private companies is less complicated than for public companies. We also find that acquirers with better governance mechanisms exhibit stronger abilities to close their deals. Consistent with Golubov *et al.* (2012), we find that tender offers have a positive effect on deal completion probability, while hostile deals and the sigma value have a negative effect on the chance of deal completion.

5.3.2. Financial hedging and deal completion time. The univariate tests in Panel D of Table 1 show that it takes FX derivatives users an average of 53 days to close a deal, while it takes non-users an average of 42 days to do so. Table 5 reports the tobit regression²² results on how an acquirer's hedging behaviour affects its deal completion time. The dependent variable is *Completion_time*, which is defined as the number of calendar days between the deal announcement date and the deal completion date. Various acquirer and deal characteristics are controlled for in the regressions. Similar to the univariate test results, we find that all the coefficients of the hedging variables are positive and statistically significant. This is consistent with Hypothesis 3, that it takes derivatives users longer to close cross-border M&A deals than non-users. Specifically, it takes an acquirer with FX hedging experience 32 days more to complete a cross-border M&A than a firm without such experience, after controlling for deal and firm characteristics.

²² In our cross-border M&A sample, some deals have a completion time of one day. The tobit regression adjusts for the left-censoring observations at one.

Table 4
Deal completion probabilities and financial hedging

This table presents the regression results of cross-border M&A deal completion probabilities on financial hedging variables. Probit (columns (1)–(4)) and logit (column (5)) regressions are estimated based on a sample of 1,369 completed and withdrawn deals carried out by S&P 1500 firms between 2000 and 2014. The dependent variable is a binary variable that takes the value of one if the deal was completed, and zero otherwise. Detailed definitions of all the variables can be found in the Appendix. Year, Fama–French 10 industry, and S&P index fixed effects are controlled for all regressions. The *p*-values are reported in parentheses. Significance at the 0.01, 0.05 and 0.10 levels is indicated by ***, ** and *, respectively.

	1	2	3	4	5
<i>Fcd</i>	−0.031 (0.804)				
<i>Fcd/Ird</i>		−0.053 (0.694)			
<i>Fcd_target</i>			0.292** (0.017)	0.322** (0.018)	0.652** (0.018)
<i>Cash</i>				0.149 (0.259)	0.283 (0.282)
<i>Equity</i>				0.221 (0.423)	0.415 (0.417)
<i>Non-public</i>				0.524*** (0.010)	0.891** (0.019)
<i>Toehold</i>				−0.001 (0.999)	−0.206 (0.864)
<i>Hostile</i>				−1.456** (0.024)	−3.052** (0.012)
<i>Tender</i>				1.112*** (0.003)	2.464*** (0.006)
<i>Related_industry</i>				−0.205 (0.115)	−0.393 (0.137)
<i>Relative_size</i>				−0.718* (0.057)	−1.484** (0.030)
<i>Size</i>				−0.044 (0.546)	−0.102 (0.481)
<i>Tobin's Q</i>				−0.037 (0.454)	−0.068 (0.490)
<i>Leverage</i>				−0.122 (0.756)	−0.123 (0.872)
<i>Cash/assets</i>				0.112 (0.832)	0.137 (0.895)
<i>Governance</i>				0.132** (0.037)	0.247** (0.048)
<i>Runup</i>				0.228 (0.249)	0.470 (0.228)
<i>Sigma</i>				−31.541***	−61.840***

Table 4
Continued

	1	2	3	4	5
<i>Foreign_debt</i>				(0.004) −0.304** (0.039)	(0.004) −0.651** (0.024)
<i>Intercept</i>	1.548*** (0.000)	1.563*** (0.000)	1.413*** (0.000)	2.099** (0.012)	3.916** (0.017)
S&P index fixed effects	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	1,369	1,369	1,219	1,171	1,171
Pseudo R-squared	0.042	0.042	0.049	0.128	0.129

Besides the hedging variables, our results are consistent with Golubov *et al.* (2012), where deals involving acquirers and targets from the same industry and deals with a large transaction value relative to acquirer size have longer completion time, while deals made by acquirers whose stocks experience high volatility prior to the deal announcement have shorter completion time. We also find that acquirers' outstanding foreign currency-denominated debt has a significantly positive effect on deal completion time. In addition, we find that tender offers have a negative effect on completion time in cross-border M&As.²³

5.4. Financial hedging and long-term performance

Our analysis so far suggests that derivatives users take longer to negotiate and complete deals, have a higher deal completion probability, and, most importantly, experience higher announcement returns than non-users. In this section, we explore whether the benefits of hedging are capitalised in acquirer long-term performance. In particular, we explore the difference in both long-term operating performance and stock performance between derivatives users and non-users.

Following Huson *et al.* (2004) and Guercio *et al.* (2008), we use changes in abnormal operating ROA ($\Delta AROA$) to measure acquirer post-merger long-term operating performance. We first compute ROA as the ratio of operating income to the book value of total assets. Then, following Barber and Lyon (1996), we adjust an acquirer's ROA by subtracting the median ROA of a control group of firms that are from the same industry – two-digit Standard Industrial Classification (SIC) code – and which have similar prior operating performance ($\pm 10\%$ ROA).²⁴ This procedure generates the

²³ Tender offers are not common in cross-border M&As. In our sample, only about 6% of the deals involve tender offers.

²⁴ When the above criteria yield no firms in the control group, we relax the requirement of the same two-digit SIC code to the same one-digit SIC code or, further, remove the industry

Table 5
Deal completion time and acquirer financial hedging

This table presents the Tobit regression results of deal completion time on acquirer financial hedging variables. The sample includes 1,282 successful cross-border M&As carried out by S&P 1500 firms between 2000 and 2014. The dependent variable, *Completion_time*, is the number of days between the deal announcement date and the acquisition effective date. Detailed definitions of all the variables can be found in the Appendix. Year, Fama–French 10 industry, and S&P index fixed effects are controlled for all regressions. The *p*-values are reported in parentheses. Significance at the 0.01, 0.05 and 0.10 levels is indicated by ***, ** and *, respectively.

	1	2	3	4
<i>Fcd</i>	32.37*** (0.000)			
<i>Fcd_target</i>		28.70*** (0.000)		
<i>Fcd/Ird</i>			22.84*** (0.007)	
<i>Hedging_scope</i>				20.28*** (0.000)
<i>Cash</i>	7.46 (0.288)	4.99 (0.524)	6.26 (0.373)	7.56 (0.282)
<i>Toehold</i>	33.55 (0.520)	37.37 (0.434)	32.98 (0.532)	25.60 (0.633)
<i>Hostile</i>	43.80 (0.294)	36.25 (0.403)	38.49 (0.408)	38.74 (0.371)
<i>Tender</i>	47.25*** (0.000)	45.41*** (0.000)	47.08*** (0.000)	46.59*** (0.000)
<i>Related_industry</i>	16.56** (0.019)	18.64** (0.017)	15.08** (0.033)	15.15** (0.032)
<i>Relative_size</i>	109.33*** (0.000)	106.07*** (0.000)	105.32*** (0.000)	108.63*** (0.000)
<i>Runup</i>	1.46 (0.894)	−0.33 (0.978)	0.59 (0.957)	0.29 (0.979)
<i>Sigma</i>	−1,940.7*** (0.001)	−1,588.8** (0.011)	−1,944.0*** (0.001)	−1,685.3*** (0.004)
<i>Foreign_debt</i>	20.63** (0.020)	24.45** (0.011)	20.97** (0.019)	16.78* (0.058)
<i>Intercept</i>	19.97 (0.456)	14.44 (0.608)	19.44 (0.482)	17.61 (0.520)
Industry fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	708	630	708	708
Pseudo R-squared	0.018	0.017	0.016	0.018

adjusted ROA (AROA), which controls for the mean reversion in the accounting performance of firms experiencing substantial corporate events or extreme performance. Lastly, we compute the change in AROA ($\Delta AROA$) over a 3-year window starting from the deal announcement.

Columns (1) to (3) of Table 6 present the multivariate regression results of hedging on acquirer long-term operating performance. We use three measures to identify acquirers' hedging programmes: *Fcd*, *Fcd/Ird* and *Hedging_scope*. After controlling for various acquirer and deal characteristics, we find all the coefficients of the three hedging measures to be positive and statistically significant. Our results show that financial hedging improves an acquirer's long-term operating performance after cross-border M&As. The impact is more pronounced for acquirers that hedge more types of risk exposure. On average, the use of FX derivatives increases acquirer $\Delta AROA$ by 0.7% over 3 years after the deal announcement.

We next examine the post-deal long-term stock performance of acquirers. As noted by Kothari and Warner (2007), risk adjustment is critically important in assessing the long-term performance of event studies and risk should be estimated based on stock performance after the event. To measure post-deal long-term stock performance, we calculate BHARs by the control firm method to reduce any bias resulting from new listings, rebalancing and skewness. Following Barber and Lyon (1997) and Lyon *et al.* (1999), for each acquirer in our sample, we identify a firm of similar size and book-to-market ratio as the benchmark. Furthermore, as pointed out by Duchin and Schmidt (2013), one of the major concerns about long-term event studies in M&As is the clustering of merger activities at the industry level, which could result in biased testing results if we assume the independence of stock returns across different firms. To mitigate this concern, we further require benchmark firms to be in the same industry as the acquirers. In summary, we match each acquirer with a firm from the same industry (two-digit SIC code), of similar size ($\pm 10\%$), and with the closest book-to-market ratio. The BHARs are calculated over a 3-year window after the deal announcement.²⁵

Columns (4) to (6) of Table 6 present the results of the long-term stock performance test. After controlling for various acquirer and deal characteristics, we find the coefficients of the three hedging measures are positive and statistically significant. Our results show that financial hedging is associated with increased acquirer long-term stock performance. The impact is more pronounced for acquirers that hedge more types of risk exposures. On average, the use of FX derivatives increases acquirer annual BHARs by 0.4% over 3 years after the deal announcement.

Overall, the results in Table 6 are consistent with Hypothesis 4 and illustrate that the benefits of hedging for acquirers extend well beyond short-term performance improvement. Over a longer horizon, both the operating performance and stock market performance of derivatives users surpass those of non-users.

6. Robustness Tests and Further Discussions

In this section, we discuss the alternative explanations of our findings and present robustness test results.

6.1. Alternative explanations and omitted variables

An acquirer's financial hedging decision may be related to omitted factors that also affect M&A outcomes. To address this concern, we investigate whether the positive

²⁵ We repeat our tests with long-term performance measured over a 4-year and a 5-year time window, respectively. The results are qualitatively the same and are available upon request.

Table 6
Long-run performance and financial hedging

This table presents the regression results of acquirer long-run performance measures on financial hedging variables. The OLS regressions with robust standard errors are based on a sample of 1,276 successful cross-border M&As made by S&P 1500 firms between 2000 and 2014. In columns (1) to (3), the dependent variable is the acquirer operating performance proxy variable $\Delta AROA$. In columns (4) to (6), the dependent variable is the acquirer buy-and-hold abnormal stock return ($BHAR$) over 3 years after the announcement date. Detailed definitions of all the variables can be found in the Appendix. Year, Fama–French 10 industry, and S&P index fixed effects are controlled for all regressions. The p -values are reported in parentheses. Significance at the 0.01, 0.05 and 0.10 levels is indicated by ***, ** and *, respectively.

	Operating Performance			Stock Performance		
	1	2	3	4	5	6
<i>Fcd</i>	0.007* (0.086)			0.004** (0.041)		
<i>Fcd/Ird</i>		0.007* (0.094)			0.007*** (0.001)	
<i>Hedging_scope</i>			0.001* (0.089)			0.003** (0.013)
<i>Cash</i>	0.002 (0.691)	0.002 (0.672)	0.002 (0.651)	-0.003** (0.044)	-0.003** (0.044)	-0.003** (0.049)
<i>Equity</i>	-0.022* (0.088)	-0.022* (0.086)	-0.021* (0.097)	-0.007* (0.069)	-0.007* (0.086)	-0.007* (0.072)
<i>Nonpublic</i>	0.017* (0.092)	0.017* (0.091)	0.017* (0.096)	-0.003 (0.174)	-0.004 (0.129)	-0.003 (0.171)
<i>Toehold</i>	0.020 (0.119)	0.020 (0.112)	0.020 (0.118)	-0.002 (0.784)	-0.003 (0.724)	-0.003 (0.719)
<i>Hostile</i>	0.009 (0.753)	0.010 (0.747)	0.010 (0.732)	-0.008 (0.582)	-0.008 (0.571)	-0.008 (0.562)
<i>Tender</i>	0.011 (0.359)	0.011 (0.341)	0.011 (0.353)	-0.002 (0.540)	-0.002 (0.450)	-0.002 (0.500)
<i>Related_industry</i>	0.004 (0.315)	0.004 (0.292)	0.004 (0.293)	0.004** (0.017)	0.004** (0.019)	0.004** (0.022)
<i>Relative_size</i>	-0.016 (0.239)	-0.015 (0.263)	-0.016 (0.244)	-0.005 (0.418)	-0.005 (0.357)	-0.005 (0.392)
<i>Size</i>	0.006** (0.019)	0.006** (0.022)	0.006** (0.032)	-0.001 (0.420)	-0.001 (0.369)	-0.001 (0.232)
<i>Tobin's Q</i>	0.007*** (0.007)	0.007*** (0.007)	0.007*** (0.007)	0.000 (0.765)	0.000 (0.708)	0.000 (0.542)
<i>Leverage</i>	0.015 (0.436)	0.014 (0.459)	0.012 (0.516)	0.004 (0.451)	0.003 (0.561)	0.003 (0.555)
<i>Cash/assets</i>	-0.015 (0.600)	-0.016 (0.553)	-0.016 (0.574)	-0.004 (0.584)	-0.003 (0.719)	-0.003 (0.679)
<i>Governance</i>	0.000 (0.910)	0.000 (0.996)	0.000 (0.965)	0.000 (0.677)	0.000 (0.548)	0.000 (0.607)

Table 6
Continued

	Operating Performance			Stock Performance		
	1	2	3	4	5	6
<i>Foreign_debt</i>	−0.005 (0.320)	−0.005 (0.323)	−0.004 (0.338)	0.002 (0.199)	0.002 (0.212)	0.002 (0.266)
<i>ROA</i>	−0.208*** (0.006)	−0.207*** (0.007)	−0.207*** (0.007)	0.000 (0.973)	−0.000 (0.967)	0.000 (0.964)
<i>Intercept</i>	−0.085*** (0.008)	−0.082*** (0.009)	−0.082*** (0.009)	0.002 (0.814)	0.001 (0.922)	0.004 (0.693)
S&P index fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	913	913	913	997	997	997
Adj R-squared	0.129	0.129	0.127	0.056	0.064	0.058

relationship between financial hedging and acquirer CARs can be explained by a series of alternative explanations.

6.1.1. Financial hedging and managerial ability. One alternative explanation for the positive effect of financial hedging on deal performance is that both hedging and deal performance are driven by managerial quality. The use of derivatives could just be the result of talented management teams and good business strategies. To examine whether ‘better managerial ability’ is the only interpretation of our results, we specifically control for managerial ability in our regressions and check the robustness of our results.

Demerjian *et al.* (2012) apply data envelopment analysis and construct a managerial ability score index based on a manager’s efficiency in transforming corporate resources to revenue. They find that this managerial ability index is strongly associated with manager fixed effects and can explain the stock price reaction to chief executive officer (CEO) turnover. We obtain managerial ability score (*Managerial_score*) data between 2000 and 2011 from Peter Demerjian’s website and include *Managerial_score* as a control variable in our analysis.

Panel A of Table 7 reports the test results of financial hedging on acquirer CARs after controlling for managerial ability. We use the endogenous binary treatment model estimated by a two-step consistent estimator, with specifications similar to those of the models shown in Table 3. Only the second-step outcome equation regression results are reported. The coefficients of *Fcd* and *Fcd/Ird* remain positive and statistically significant in all the outcome equations, suggesting that the better managerial ability story cannot fully account for our findings. The coefficient of *Managerial_score* is statistically significant but negative. An explanation consistent with this result is the hubris hypothesis (e.g., Roll, 1986; Malmendier and Tate, 2005), that good past performance tends to induce CEO hubris so that acquirers choose improper targets or overpay a deal premium, leading to inferior deal performance.

Table 7

Acquirer announcement returns and financial hedging: Alternative explanations

This table presents the binary treatment model regression results of acquirer CARs on financial hedging variables, with control variables for alternative explanations. We only report the results of the second-step outcome equations. The first-step treatment regression results are similar to those reported in Table 3. The first-step treatment equation is estimated by a probit regression, where the dependent variable is *Fcd* (columns (1) and (2)) and *Fcd/Ird* (columns (3) and (4)). In columns (1) and (3), the instrument variable used in the treatment equation is *Analyst_number*. In columns (2) and (4), the instrument variable used in the treatment equation is *Foreign_sales/Sales*. In the outcome equations, the dependent variable is acquirer CARs estimated by the market model. All the control variables that are not listed separately in the outcome equations are the same as those in Table 3. Year, industry, and S&P index fixed effects are controlled for all regressions. The *p*-values are reported in parentheses. Significance at the 0.01, 0.05 and 0.10 levels is indicated by ***, ** and *, respectively.

Panel A. Managerial ability. We control for managerial ability variable, *Managerial_score* (Demerjian *et al.*, 2012), in the outcome equations. This index is obtained from Peter Demerjian's website, which is then matched to our cross-border M&A sample over the period 2000 to 2011.

	1	2	3	4
<i>Fcd</i>	0.137*** (0.004)	0.038* (0.053)		
<i>Fcd/Ird</i>			0.122*** (0.000)	0.074*** (0.002)
<i>Managerial_score</i>	-0.042** (0.026)	-0.042** (0.018)	-0.047** (0.012)	-0.048*** (0.006)
Control variables	Yes	Yes	Yes	Yes
S&P/Industry/Year fixed effects	Yes	Yes	Yes	Yes
Observations	949	1,051	949	1,051
<i>Prob > chi2</i>	0.000	0.000	0.000	0.000

Panel B. Foreign operation exposure. We control for the number of mentions of target nations in acquirers' 10-K reports, *Counts*, in the outcome equations.

	1	2	3	4
<i>Fcd</i>	0.106** (0.014)	0.042* (0.065)		
<i>Fcd/Ird</i>			0.090*** (0.008)	0.066** (0.014)
<i>ln(Counts)</i>	0.000 (0.826)	0.000 (0.971)	0.000 (0.890)	0.000 (0.929)
Control variables	Yes	Yes	Yes	Yes
S&P/Industry/Year fixed effects	Yes	Yes	Yes	Yes
Observations	1,072	1,174	1,072	1,174
<i>Prob > chi2</i>	0.000	0.000	0.000	0.000

Table 7
Continued

Panel C. Serial acquirers. We control for serial acquirers in the outcome equations. The variable *Serial_acquirer* is equal to the number of deals carried out by the same acquirer before the deal announcement in our sample

	1	2	3	4
<i>Fcd</i>	0.110*** (0.009)	0.034* (0.062)		
<i>Fcd/Ird</i>			0.105*** (0.001)	0.066*** (0.003)
<i>Serial_acquirer</i>	-0.001 (0.594)	0.000 (0.919)	-0.001 (0.535)	0.000 (0.787)
Control variables	Yes	Yes	Yes	Yes
S&P/Industry/Year fixed effects	Yes	Yes	Yes	Yes
Observations	1,081	1,184	1,081	1,184
<i>Prob > chi2</i>	0.000	0.000	0.000	0.000

Panel D. Commodity. We control for CD usage in the outcome equations. The variable *Commodity* equals one if an acquirer uses CDs in the fiscal year prior to the deal announcement, and zero otherwise.

	1	2	3	4
<i>Fcd</i>	0.102** (0.015)	0.031* (0.085)		
<i>Fcd/Ird</i>			0.098*** (0.003)	0.061*** (0.008)
<i>Commodity</i>	-0.008 (0.267)	-0.011* (0.083)	-0.006 (0.366)	-0.009 (0.180)
Control variables	Yes	Yes	Yes	Yes
S&P/Industry/Year fixed effects	Yes	Yes	Yes	Yes
Observations	1,081	1,184	1,081	1,184
<i>Prob > chi2</i>	0.000	0.000	0.000	0.000

Panel E. All four variables. We add all four control variables from Panels A to D together in the outcome equations.

	1	2	3	4
<i>Fcd</i>	0.144*** (0.005)	0.045* (0.078)		

Table 7
Continued

Panel E. All four variables. We add all four control variables from Panels A to D together in the outcome equations.

	1	2	3	4
<i>Fcd/Ird</i>			0.125*** (0.001)	0.078*** (0.010)
<i>Managerial_score</i>	-0.052** (0.015)	-0.055*** (0.007)	-0.057*** (0.008)	-0.061*** (0.003)
<i>ln(Counts)</i>	0.001 (0.566)	0.000 (0.883)	0.001 (0.606)	0.000 (0.934)
<i>Serial_acquirer</i>	-0.001 (0.626)	0.000 (0.905)	-0.001 (0.639)	0.000 (0.863)
<i>Commodity</i>	-0.004 (0.675)	-0.010 (0.199)	-0.001 (0.920)	-0.007 (0.419)
Control variables	Yes	Yes	Yes	Yes
S&P/Industry/Year fixed effects	Yes	Yes	Yes	Yes
Observations	718	795	718	795
<i>Prob > chi2</i>	0.000	0.000	0.000	0.000

6.1.2. *Acquirer CARs and foreign operating exposures.* If an acquirer has foreign operations in the target firm's country before the M&A deal, it may be motivated to hedge its FX risk related to its foreign operating exposures. On the other hand, an acquirer with target country operations experience could also make better cross-border M&A decisions. Thus, it might be the acquirer's knowledge of foreign markets rather than the financial hedging itself that leads to the improved deal performance documented in our empirical tests. In this section, we address the concern that omitting an acquirer's pre-deal operating exposures in the target nation could introduce an estimation bias in our empirical analysis.

Since foreign sales and foreign production information at the country level is not disclosed by the acquirers, we adopt an alternative approach to measure their foreign exposures. Garcia and Norli (2012) extract state name counts from 10-K reports and use the counts as a measure of the degree of geographic dispersion of firm business operations. Following their approach, we count the number of mentions of foreign target nations (*Counts*) in acquirer 10-K reports prior to the deal announcement.²⁶ The variable *Counts* is used as a proxy for an acquirer's foreign operations experience and knowledge of the target nation before the M&A deal.

Panel B of Table 7 presents the test results of financial hedging on acquirer CARs after controlling for acquirer foreign operations experience. We use the endogenous binary treatment model estimated by a two-step consistent estimator. Only the second-step outcome equation regression results are reported. The coefficient of *Counts* is found to be

²⁶ The variations, including official names, unofficial names, and their adjectival forms, are all considered when we count the mentions of target nations.

statistically insignificant in our results. The coefficients of *Fcd* and *Fcd/Ird* remain positive and statistically significant in all the outcome equations.

6.1.3. Acquirer CARs and serial acquirers. Some deals in our sample are carried out by serial acquirers. If the improvement in acquirer CARs is driven by frequent acquirers that are also more likely to use derivatives, then the conclusion of our empirical results is compromised. In our sample of 1,369 cross-border M&A deals, there are 639 unique acquirers, about half of which (316) carry out a single transaction in our sample. A group of 55 frequent acquirers carry out at least five deals each. To address concerns that our results are driven by these frequent acquirers, we repeat our multivariate tests after dropping the deals made by these repeat acquirers from our sample. In untabulated test results, we find that hedging variables remain positively related to deal performance. Apart from the concern discussed above, recent M&A studies also demonstrate that, if an acquirer makes a series of acquisitions, the acquirer CARs could decline as it conducts more deals (e.g., Billett and Qian, 2008; Aktas *et al.*, 2011). We control for the variable *Serial_acquirer* in the outcome equation of the endogenous binary treatment model, defined as the number of deals carried out by the same acquirer before the deal announcement in our sample. As shown in Panel C of Table 7, our results are robust after controlling for the serial acquirer effect.

6.1.4. Acquirer CARs and CDs. Besides FCDs and IRDs, companies could also use CDs for hedging purposes. Empirical studies on commodity price hedging usually examine firms in industries with strong risk exposure to commodity prices.²⁷

One alternative interpretation of our empirical findings on the positive relationship between acquirer CARs and financial hedging is that the better deal performance is driven by unobserved firm qualities that lead firms to hedge, instead of being driven by hedging practice itself. In other words, financial hedging may have no causal effect on deal performance but only signals unobserved firm qualities. If this explanation is true, then the use of CDs should have similar correlations with acquirer announcement returns as the use of FCDs or IRDs.

To check this alternative interpretation, we add *Commodity* as a control variable in the outcome equation of the endogenous binary treatment model. The variable *Commodity* is a binary variable equal to one if an acquirer reports using CDs, and zero otherwise. Panel D of Table 7 shows that the coefficients of *Fcd* and *Fcd/Ird* remain positive and statistically significant in all the outcome equations. The coefficients of *Commodity* in the outcome equations are mostly negative and statistically insignificant. In untabulated tests, we examine the impact of commodity price hedging on acquirer CARs without controlling for FCDs and IRDs. We find that the CDs alone do not contribute to better deal performance. These findings alleviate the concern that financial hedging only signals hidden firm characteristics.

Finally, we include all four control variables, *Managerial_score*, *Counts*, *Serial_acquirer* and *Commodity*, together in the multivariate regressions shown in Panel E of Table 7. We continue to find that financial hedging is positively related to acquirer CARs.

²⁷ For example, Carter *et al.* (2006) study the use of jet fuel derivatives in a sample of US airline companies and find that commodity price hedging is positively related to Tobin's Q. By focusing on a sample of oil refiners, MacKay and Moeller (2007) also find similar results.

Table 8

Acquirer announcement returns and financial hedging: The triple fixed effects model

This table presents the regression results of acquirer CARs on financial hedging variables with a triple fixed effect estimation. The regressions are based on a sample of 1,274 completed deals carried out by S&P 1500 firms between 2000 and 2014. Following Gormley and Matsa (2014), we use the triple interaction fixed effects model (Year $\times$ Acquirer Fama–French 10 industry $\times$ Target Fama–French 10 industry) to control for unobserved acquirer and target characteristics. The dependent variable is the acquirer CAR over the 11-day event window $(-5, +5)$, where day 0 is the announcement day. The benchmark is estimated by the market model with the CRSP value-weighted index over the pre-announcement window $(-300, -91)$. The control variables are the same as those in Table 2. Detailed definitions of all the variables can be found in the Appendix. The p -values are reported in parentheses. Significance at the 0.01, 0.05 and 0.10 levels is indicated by ***, ** and *, respectively.

	1	2	3	4	5	6	7
<i>Fcd</i>	0.012** (0.015)	0.010** (0.036)	0.010** (0.044)	0.010** (0.044)			
<i>Fcd/Ird</i>					0.011** (0.043)		
<i>Hedging_scope</i>						0.005 (0.150)	
<i>Nv_derivatives</i>							0.015** (0.039)
Control variables	No	Yes	Yes	Yes	Yes	Yes	Yes
Triple fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,262	1,262	1,215	1,215	1,215	1,215	1,061
Adj R-squared	0.065	0.076	0.083	0.083	0.083	0.082	0.090

6.1.5. *Acquirer CARs and financial hedging: Unobservable firm characteristics.* To further control for unobserved acquirer or target characteristics that could affect both financial hedging and acquirer CARs, we follow Gormley and Matsa (2014) and control for the triple fixed effects of Year $\times$ Acquirer Fama–French 10 industry $\times$ Target Fama–French 10 industry.²⁸ We re-estimate all seven OLS regressions in Table 2 with the triple fixed effects model. The results in Table 8 show that our empirical results presented in Table 2 remain robust.

6.2. *Alternative endogeneity control method: PSM*

In Section 5.2, we apply the Heckman selection model to control for endogeneity concerns due to self-selection bias and unobserved firm characteristics. As an alternative approach, we follow Bartram *et al.* (2011) and employ a self-selection control method, PSM, by matching the derivatives users with a group of non-users with similar pre-acquisition firm characteristics.

²⁸ We thank Todd A. Gormley for providing the computer code of the high-dimensional fixed effect model at www.kellogg.northwestern.edu/faculty/matsa/htm/fe.htm.

For each acquirer, we generate a propensity score that is estimated by a logit model. The dependent variable in the logit model is *Fcd* and the independent variables include *Leverage*, *Cash/assets*, *Governance*, *Tobin's Q* and *Relative_size*. We choose these independent variables because Caliendo and Kopeinig (2008) argue that the independent variables in the first-step regression should strongly affect the implementation of the financial hedging programme and at least weakly affect the outcome variables. Then we match deals carried out by derivatives users with those carried out by a group of non-users based on their propensity scores. Both nearest-neighbour matching and Gaussian kernel matching methods are applied. The differences in deal outcomes between derivatives users and non-users are then estimated and reported in Table 9. Our findings in Section 5 remain robust with the PSM approach.

6.3. Currency volatility and announcement returns

Market uncertainty is critically influential on M&A activities. In cross-border M&As, high currency volatility introduces great uncertainty for the deals. High currency volatility not only increases an acquirer's payment risk during the interim period, but also increases its subsequent cash flow risk and balance sheet risk. We therefore expect that financial hedging will have a greater impact on deal performance during a high exchange rate volatility regime than during a low exchange rate volatility regime.

In online Appendix C1, we use the standard deviations of 12 monthly trade-weighted US dollar index returns prior to the deal announcement and the standard deviations of 12 monthly returns of exchange rates between the US dollar and the target nation's currency prior to the deal announcement as two measures of foreign currency volatility. We then run OLS regressions of acquirer CARs on acquirer financial hedging proxy

Table 9
Propensity score-matched acquisition outcomes

This table reports the cross-border M&A deal outcomes adjusted using propensity scores that are estimated by logit regressions of the FX hedging likelihood on firm characteristics. The sample includes 865 deals announced by FX derivatives users and 504 deals announced by non-FX derivatives users. The independent variables in the logit regressions are *Relative_size*, *Leverage*, *Cash/assets*, *Governance* and *Tobin's Q*. The differences in deal outcomes between deals announced by FX derivatives users (treated sample) and propensity score-matched deals announced by non-FX derivatives users (control sample) are reported. Detailed definitions of all the variables can be found in the Appendix. Significance at the 0.01, 0.05 and 0.10 levels is indicated by ***, ** and *, respectively.

	10 Nearest	30 Nearest	50 Nearest	Gaussian Kernel
<i>CAR</i> _[−5,5]	0.009* (0.076)	0.008* (0.078)	0.008** (0.044)	0.010 (0.150)
<i>Completion</i>	0.039** (0.026)	0.042** (0.018)	0.042** (0.017)	0.052* (0.063)
<i>Completion_time</i>	11.4 (0.133)	12.0* (0.088)	7.9 (0.408)	20.1*** (0.000)
$\Delta ROA_{t,t+3}$	0.008** (0.045)	0.010*** (0.001)	0.012*** (0.002)	0.011*** (0.065)

variables over high currency volatility and low currency volatility periods, respectively. We find that the coefficients for all the financial hedging measures are positive and statistically significant when the deals are announced following a period of high currency volatility, but the coefficients become positive yet statistically insignificant when they are announced after a period of low currency volatility. These findings lend support to the notion that financial hedging improves acquirer performance through risk reduction and the effect is more significant during volatile times.

6.4. *Other hedging methods*

Besides hedging with derivatives, financial hedging with foreign currency debt and operational hedging are also commonly used by firms to reduce their foreign currency exposures. A firm with foreign currency risk exposures could finance its operations in the local nation's currency to create a natural hedge (Graham and Harvey, 2001). This approach is commonly used by firms with long-term exposures of foreign currency risk (Clark and Judge, 2016). In our main tests, we include foreign currency-denominated debt as a control variable and the coefficients of our hedging proxies are statistically significant. This is not surprising, given that a portion of risk exposures associated with cross-border M&As are in the near future and of a predictable amount, which can most effectively be managed by financial derivatives hedging. Our findings suggest that it is beneficial for acquirers to hedge these risk exposures with financial derivatives, even if certain acquirers may adopt a natural hedge through borrowing in local currencies as a complement.

Operational hedging refers to the strategy where a firm relocates its production facilities or shifts its source of inputs to match its costs and revenues in the same currency. Kim *et al.* (2006) find a substitute relationship between financial hedging and operational hedging for non-financial firms. Hankins (2011) finds that, for a sample of firms in the US financial industry, M&As provide operational hedging by reducing the volatility of the combined entity's operational income. We cannot rule out that certain cross-border acquisitions in our sample are for operational hedging purposes. If this is the case, then the marginal benefits of derivatives hedging will be smaller. The fact that we find a significant effect of financial hedging on deal performance, given that some deals may be carried out for operational hedging purposes, only makes our findings more significant.

6.5. *Target country fixed effects*

In our sample, an acquirer may have different risk exposures in different target countries and the difference is not random. Therefore, it may be necessary to control for target country fixed effects. The tests in Section 6.3 are a sub-sample analysis based on country-to-country exchange rate volatilities, which partially addresses target country fixed effects. To further address this issue, we re-estimate the OLS regressions in Panel A of Table 2 and control for target country fixed effects. The test results remain qualitatively the same as reported in Panel A of Table 2.²⁹

²⁹ The untabulated test results are available upon request.

6.6. Financial hedging and acquirer stock return volatility

High stock return volatility leads to higher external financing costs and more uncertainty on payments if the method of payment includes stock swaps. Using a large sample of international firms, Bartram *et al.* (2011) find strong evidence that the use of financial derivatives reduces both total risk and systematic risk. We examine the effect of hedging on stock return volatility in our sample as a performance test.

The stock return volatility measure we adopt is the implied volatility at the deal announcement, which reflects the market's expectation of an acquirer's future stock volatility. Following Duchin and Schmidt (2013), we collect acquirer implied volatility data from the estimated volatility surface in the OptionMetrics database for ATM options with a time to maturity of 91 days³⁰ and calculate the average implied volatility of ATM call and put options. In online Appendix C2, we use both univariate and multivariate tests to show that derivatives users have lower market perceived future stock return uncertainty than non-users upon announcement dates. On average, hedging with FX derivatives reduces an acquirer's 91-day implied volatility by 0.047 on the announcement date.

6.7. Financial hedging reported after the deal announcement

The financial hedging data employed in this study are collected from the acquirer's annual report filed prior to the deal announcement. This practice is consistent with the standard event study approach in M&A literature and mitigates concerns of reverse causality. However, some deal-specific hedging activities by acquirers could occur during the fiscal years when the M&As are announced and are subsequently disclosed in the 10-K reports available after the deal announcement. In this case, the financial hedging information reported after the deal announcement would also be of interest. In addition, given that deal-specific hedging information is not disclosed by all acquirers, changes in financial hedging activities reported in the two consecutive annual reports before and after deal announcement can offer insight on deal-related hedging activities. To further explore the link between hedging programmes and cross-border M&A activities, we manually collect acquirers' financial hedging data from their annual reports filed for the fiscal years in which the deals are announced.³¹

In untabulated tests, the means of *Fcd*, *Fcd_target*, *Hedging_scope*, *Nv_derivatives* and *Foreign_debt* are found to be significantly higher in the annual reports filed after the deal announcement than those filed before the deal announcement, suggesting that, in anticipation of deal completion, acquirers increase their FX hedging activities around the deal announcement. Hedging activities related to IR risk and commodity price risk do not increase significantly over the 2-year period. Furthermore, we find that financial hedging activities reported after the deal announcement have a similar effect on acquirer announcement returns as those reported before the deal announcement. The results in Table 3 are robust when we use *Fcd* and *Fcd/Ird* reported after the deal announcement as the independent variable for hedging. Finally, we calculate the change in hedging

³⁰ We also collect data for options with a time to maturity of 30 days and 60 days. Our test results are qualitatively the same and are available upon request.

³¹ We thank an anonymous referee for suggesting this analysis.

activities between the two fiscal years ($\Delta Hedging_scope$ and $\Delta Nv_derivatives$) and find that they both have a positive effect on acquirer announcement returns.

7. Conclusions

Using hand-collected data on reported derivatives information from S&P 1500 firms that acquire foreign targets between 2000 and 2014, we empirically examine whether financial hedging affects cross-border M&A deal performance. Our sample acquirers encounter an increase in FX risk as well as in IR risk if a deal needs external financing. Using an event study approach, we find that derivatives users have higher announcement returns than non-users. The improvement in acquirer CARs is both statistically and economically significant. Over a time window of $(-5, +5)$ around the deal announcement, FCD users experience an average of 1% higher abnormal returns than non-users. Besides improved acquirer CARs, we show that financial hedging correlates with higher deal completion probabilities, longer deal completion time, and better long-term acquirer performance.

Overall, our results are consistent with the optimal hedging theory that financial hedging improves firm value in the presence of market frictions. To address endogeneity concerns, we use the Heckman treatment effect model to control for hidden variables that could be correlated with both hedging decisions and deal performance. In our robustness tests, we apply both PSM and the triple fixed effects models to further mitigate potential endogeneity. To rule out the alternative explanations of our results, we conduct a series of robustness tests to directly control for specific firm or deal characteristics, including firm size, stock return momentum, managerial ability, foreign operating exposures, repeated acquirers, and target country fixed effects. Our results remain robust after we control for these variables. Adding commodity price hedging as a control variable, we find that, unlike FX and IR hedging, commodity hedging does not correlate with better deal performance. This lends support to the notion that hidden firm characteristics motivating firms to hedge cannot fully explain the improved deal performance. Furthermore, by separating our sample period into a low exchange rate regime and a high exchange rate regime, we confirm that hedging-related acquirer performance improvement is more pronounced during a high currency volatility regime than during a low currency volatility regime. This reinforces the notion that hedging benefits firms through risk exposure reduction. Lastly, we show that the implied volatility for derivatives users is lower than for non-users during the deal announcement window. The observed risk reduction effect is consistent with the results of Bartram *et al.* (2011).

Our findings contribute to the literature on the link between hedging and firm value improvement. To our knowledge, this is the first study to directly examine the relationship between hedging and firm investment performance. Similar to previous M&A studies, we only observe and examine deals that are made public. Boone and Mulherin (2007) find that public takeover activities are only the tip of the iceberg of total takeover activities. It is possible that some deals are not made public due to the lack of acquirer capacity to manage deal-related risks, which further emphasises the importance of financial hedging in cross-border M&As. Current US financial statement reporting standards do not require firms to disclose deal-specific hedging information. Therefore, we follow the practice of the empirical hedging studies and collect general hedging information at the firm level. If changes in financial reporting standards make more detailed corporate hedging data available in the future, we would be able to not only

study the relationship between hedging and cross-border M&As more thoroughly but also examine hedging's impact on other types of firm operations.

Appendix: Variable definitions

This table provides variable definitions and the corresponding data sources. In this table, CRSP refers to the Center for Research in Security Prices, FF refers to Kenneth French's website at Dartmouth, SDC refers to Thomson Reuters Securities Data Company, IBES refers to the Institutional Brokers Estimate System, ISS refers to the Institutional Shareholder Services (formerly RiskMetrics), and EDGAR refers to the SEC's Electronic Data Gathering, Analysis, and Retrieval system.

Variable	Definition	Source
Deal outcomes		
<i>CAR</i> _[X, Y]	CARs over the event window [X, Y] days surrounding the acquisition announcement, using the market model with the CRSP value-weighted index.	CRSP
<i>Completion</i>	1 for completed deals, 0 for withdrawn deals.	SDC/LexisNexis
<i>Completion_time</i>	Number of days between announcement and effective dates.	SDC/LexisNexis
<i>AROA</i>	Acquirer adjusted return on assets (AROA) with the benchmark being the median ROA of a group of Compustat firms within the same two-digit SIC code and similar operating performance ($\pm 10\%$ ROA) before the deal announcement.	Compustat
<i>BHAR</i> (control firm)	Acquirer buy-and-hold abnormal stock return with the benchmark being the return of a <i>control_rm</i> with the same 2-digit SIC code, similar size ($\pm 10\%$), and the nearest book-to-market ratio.	Compustat/CRSP/ FF Data Library
<i>Implied volatility</i>	The implied volatility of acquirer stock returns at the deal announcement. It is the average implied volatilities of the ATM call option and the ATM put option with the same maturity.	Option Metrics
Deal characteristics		
<i>Cash</i>	1 for deals financed fully with cash, 0 otherwise.	SDC
<i>Equity</i>	1 for deals financed partially or fully with stock, 0 otherwise.	SDC
<i>Non-public</i>	1 if the target is not a public firm, 0 otherwise.	SDC
<i>Toehold</i>	1 if the acquirer has already held a certain percentage of the target shares at the announcement, 0 otherwise.	SDC
<i>Hostile</i>	1 for hostile deals, 0 otherwise.	SDC
<i>Tender</i>	1 for tender offers, 0 otherwise.	SDC
<i>Related_industry</i>	1 if the target and the acquirer have the same two-digit SIC code, 0 otherwise.	SDC
<i>Relative size</i>	The ratio of transaction value to acquirer market value at the end of the fiscal year before the deal is announced.	SDC/Compustat
<i>Transaction_value</i>	Value of the transaction, in millions of dollars.	SDC

Continued		
Variable	Definition	Source
Firm characteristics		
<i>Size</i>	Natural log of the acquirer's market value, adjusted for inflation.	Compustat
<i>Tobin's Q</i>	Acquirer's Tobin's Q at the end of the fiscal year before the deal is announced, following Baker and Wurgler (2002).	Compustat
<i>Leverage</i>	Acquirer's ratio of the book value of debt to the book value of total assets at the fiscal year-end before the deal is announced.	Compustat
<i>Cash/assets</i>	Acquirer's cash, normalised by total assets.	Compustat
<i>Governance</i>	We construct a corporate governance index based on six provisions (Bebchuk <i>et al.</i> , 2009). Each acquirer is given a score, from zero to six, based on the number of these provisions that the company has before the announcement: staggered boards, limits to shareholder bylaw amendments, poison pills, golden parachutes, and supermajority requirements for mergers and charter amendments.	ISS
<i>Runup</i>	Market adjusted buy-and-hold return of the acquirer's stock over a (−205, −6) window (Golubov <i>et al.</i> , 2012).	CRSP
<i>Sigma</i>	Standard deviation of the acquirer's market-adjusted daily return over a (−205, −6) window (Golubov <i>et al.</i> , 2012).	CRSP
<i>Assets</i>	Book value of acquirer total assets.	Compustat
<i>Analyst number</i>	Number of analysts following the acquirer in the fiscal year preceding the deal announcement (Geczy <i>et al.</i> , 1997).	IBES
<i>Foreign sales/Sales</i>	The ratio of the acquirer's foreign sales over its total sales at the end of the fiscal year preceding the deal announcement.	Compustat Segments
<i>ROA</i>	Operating income before depreciation divided by total assets.	Compustat
<i>Managerial score</i>	A measure of managerial ability based on managers' efficiency in generating revenues (Demerjian <i>et al.</i> , 2012).	Peter Demerjian's website
<i>Counts</i>	The number of mentions of the target country in the acquirer's 10-K report prior to the deal announcement.	EDGAR 10-K
<i>Serial acquirer</i>	The number of deals carried out by the same acquirer before the deal announcement in our sample.	SDC
Acquirer financial hedging characteristics		
<i>Fcd</i>	1 if the acquirer uses FCDs in the fiscal year before the deal announcement, 0 otherwise.	EDGAR 10-K
<i>Fcd target</i>	1 if the acquirer uses the target nation's currency derivatives in the fiscal year before the deal announcement, 0 otherwise.	EDGAR 10-K

Continued

Variable	Definition	Source
<i>Ird</i>	1 if the acquirer uses IRDs in the fiscal year before the deal announcement, 0 otherwise.	EDGAR 10-K
<i>Fcd/Ird</i>	1 if the acquirer uses either FCDs or IRDs in the fiscal year before the deal announcement, 0 otherwise.	EDGAR 10-K
<i>Hedging_scope</i>	2 if the acquirer uses both of the two types of derivatives contracts (FX and IR) in the fiscal year before the deal announcement, 1 if the acquirer uses only one of the two types of derivatives contracts (FX or IR), 0 if the acquirer does not use FCDs or IRDs.	EDGAR 10-K
<i>Nv_derivatives</i>	Notional value of the financial derivatives contracts held by the acquirer at the end of the fiscal year before the deal announcement, normalised by the acquirer's total assets.	EDGAR 10-K
<i>Foreign debt</i>	1 if the acquirer uses debt denominated in foreign currencies in the fiscal year before the deal announcement, 0 otherwise.	EDGAR 10-K/ SDC
<i>Global New Issues</i>		
<i>Commodity</i>	1 if the acquirer uses CDs in the fiscal year before the deal announcement, 0 otherwise.	EDGAR 10-K

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Supporting Information

Additional supporting information may be found in the online version of this article at the publisher's website:

Appendix A: The Statements of Financial Accounting Standard (SFAS) and the sample period selection

Appendix B: The selected examples of US acquirers hedging the financial risks associated with the cross-border deals

Appendix C Table C1: Financial hedging, CARs, and currency volatility

Table C2: Acquirer stock return volatility around cross-border M&A announcements